

Field Service WorkOrder Optimization

Name of the Student	Team ID	Email
Peddapalem Balu	LTVIP2025TMID29619	balub0115@gmail.com
Annem Karthik	LTVIP2025TMID29619	karthikleo765@gmail.com
Nandyala Munisankar	LTVIP2025TMID29619	munisankar957@gmail.com
Arigala Sagar	LTVIP2025TMID29619	sagararigala@gmail.com

Field service workorder optimization



INTRODUCTION

The Field Service Work Order Optimization System streamlines operations for a company providing installations and repairs. Utilizing a robust database, the system efficiently matches work orders with skilled technicians based on technician's location, availability, and skills. The system employs a prioritization algorithm, focusing on assigning tasks to technician. Automated communication keeps technicians informed, while analytics offer insights for continuous improvement. Overall, this solution maximizes efficiency, reduces operational costs, and improves customer satisfaction in the dynamic realm of field service operations.

Field service workorder optimization

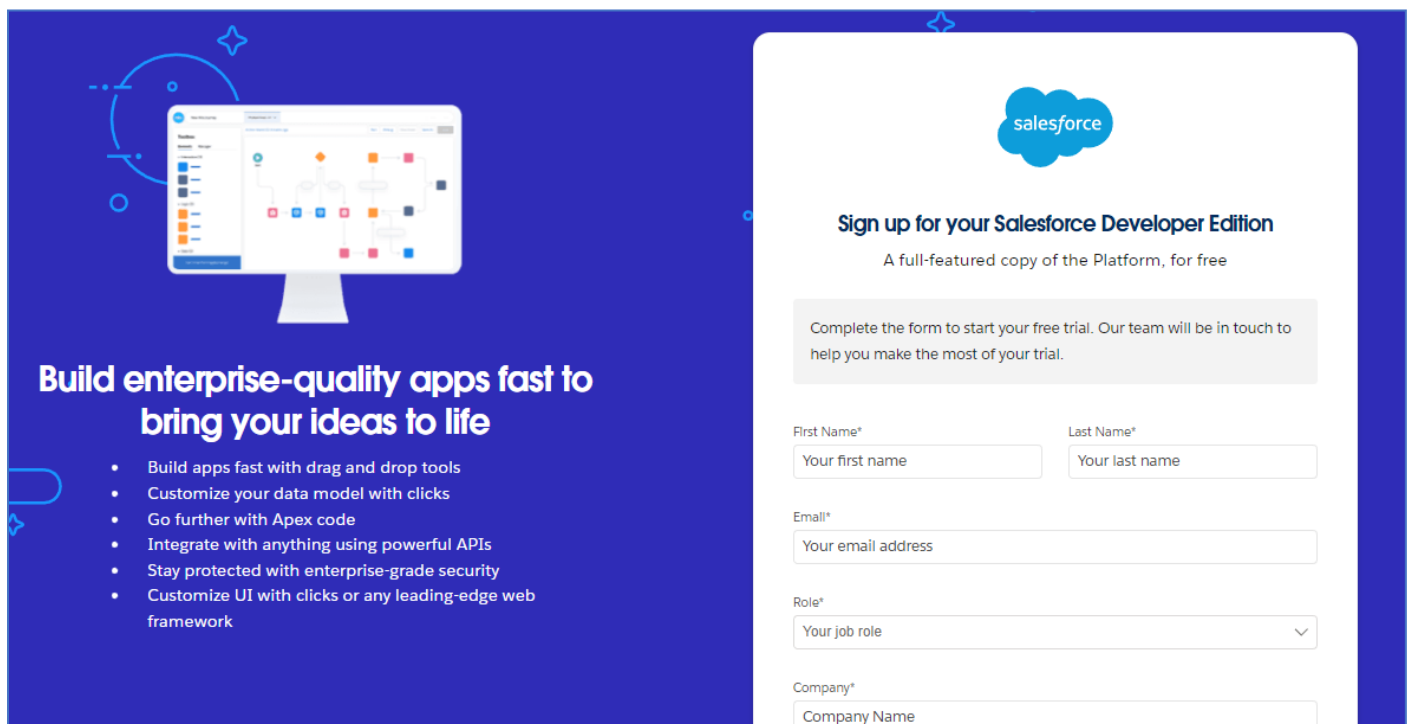
Milestone – 1

Sales force

Activity 1: Creating Developer Account

Creating a developer org in sales force.

1. Go to <https://developer.salesforce.com/signup>
2. On the sign up form, enter the following details:



Build enterprise-quality apps fast to bring your ideas to life

- Build apps fast with drag and drop tools
- Customize your data model with clicks
- Go further with Apex code
- Integrate with anything using powerful APIs
- Stay protected with enterprise-grade security
- Customize UI with clicks or any leading-edge web framework

Sign up for your Salesforce Developer Edition
A full-featured copy of the Platform, for free

Complete the form to start your free trial. Our team will be in touch to help you make the most of your trial.

First Name*

Last Name*

Email*

Role*

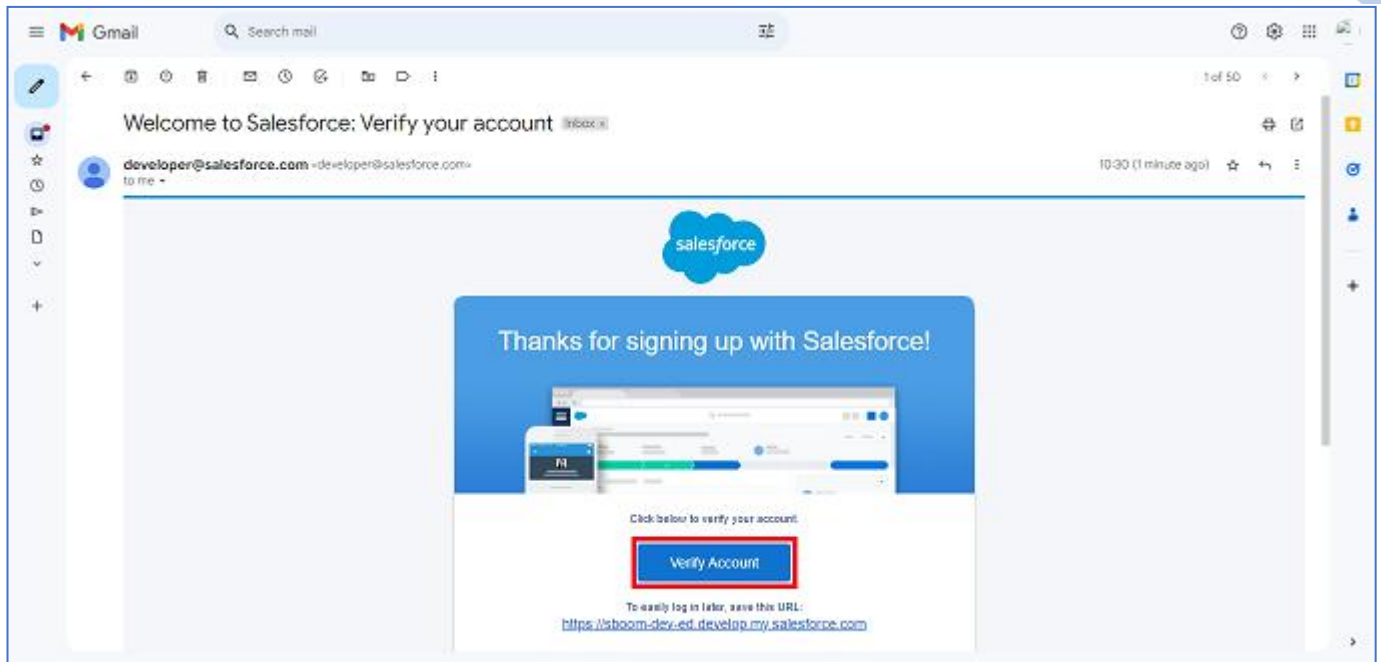
Company*

1. First name - BALU
2. Last name - PEDDAPALEM
3. Email - balub0115@gmail.com
4. Role: Developer
5. Company: Gayatri Degree College Tirupati
6. County: India

ACTIVITY 2: Account Activation

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.

Field service workorder optimization

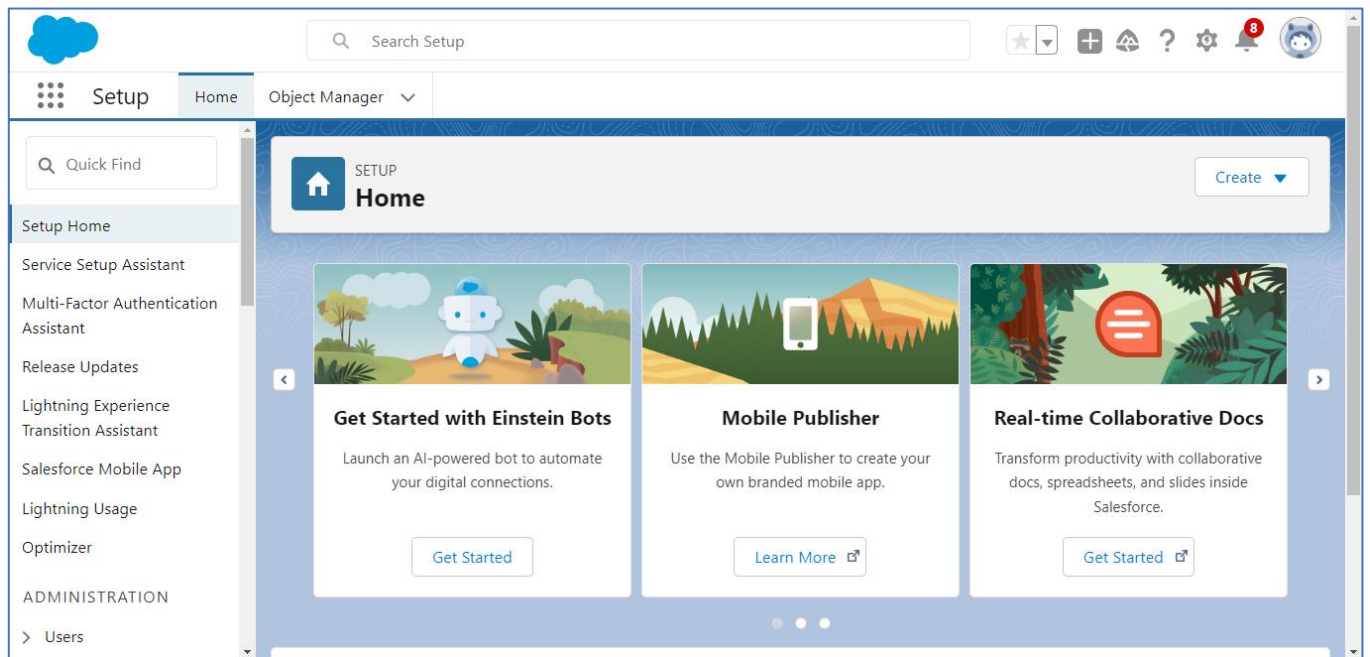


2. Click on Verify Account
3. Give a password and answer a security question and click on change password.

A screenshot of the 'Change Your Password' form. The form is titled 'Change Your Password' and asks to 'Enter a new password for lead@sb.com'. It lists requirements: 8 characters, 1 letter, and 1 number. The form contains four input fields: 'New Password' (with a 'Good' status), 'Confirm New Password' (with a 'Match' status), 'Security Question' (a dropdown menu showing 'In what city were you born?'), and 'Answer' (with the text 'asdfghjkl'). A 'Change Password' button is at the bottom, all of which are enclosed in a red rectangular border.

4. Then you will redirect to your sales force setup page.

Field service workorder optimization



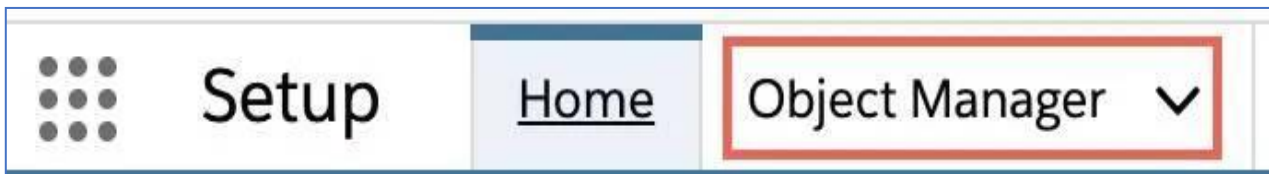
Field service workorder optimization

Milestone – 2

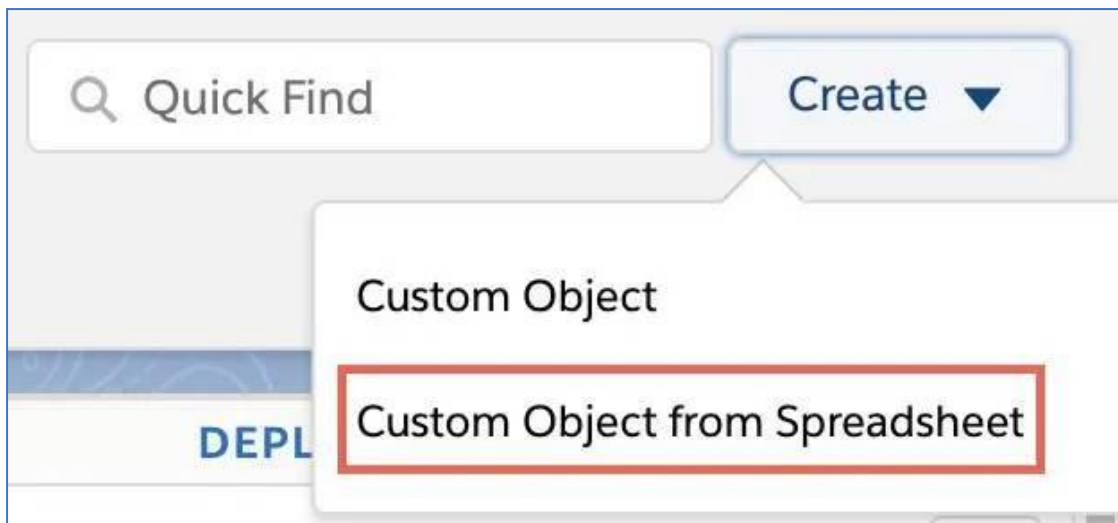
Object

Activity - 1: Create Technician Object

1. Download and open [this spreadsheet](#), edit the email column (provide your email for at least one or two records) and save it as Technician.csv.
2. Log into your salesforce account, click on Gear icon, then select Setup.
3. Click the Object Manager tab.



4. Click Create.
5. Select Custom Object from Spreadsheet.



6. Click Login with Salesforce.
7. Enter your Salesforce account username and password. (which you have created in the Milestone 1, Activity 1)
8. Click Log In.
9. Click Allow.
10. Click Upload.
11. Navigate to the Technician.csv file you downloaded and upload it. Salesforce automatically detects the fields and populates all its record data. Choose Technician ID as the Record Name field and make sure all fields are with the proper datatypes as below as they are.

Field service workorder optimization

CSV File Details

Encoding Format: Values Separated By: Field Label Source: ☐ Enter manually ☒ Detect from row * Field Labels Row: Import 5 rows of Data?: ☐ No, skip import ☒ Yes, import data Record Name Field:

Fields 7 of 7 to import ☐ Hide mapped fields

IMPORT FILE FIELD NAME	SALESFORCE FIELD NAME	SALESFORCE FIELD TYPE	ADD TO LAYOUTS	FIELD PREVIEW
✓ Technician ID	✕ Technician ID	Text	☒	T-0001
✓ Name	✕ Name	Text	☒	Raghu
✓ Phone	✕ Phone	Phone	☒	7892341560
✓ Email	✕ Email	Email	☒	example@gmail.com
✓ Location	✕ Location	Picklist	☒	Hyderabad
✓ Availability	✕ Availability	Picklist	☒	Available
✓ Skills	✕ Skills	Picklist	☒	Machine Installation

Back ☒ ☐ Next

12. Click Next and enter the following settings.

13. Click Finish. The Technician object is successfully created and data imported, all within minutes.

Field service workorder optimization

ACTIVITY 2: Create Work Order Object

Create work Order object, just as we have created Technician Object using [this spreadsheet](#) :

Note: Make sure you do field mapping with proper field type as shown below.

CSV File Details

Encoding Format: Unicode (UTF8) | Values Separated By: Comma | Field Label Source: Detect from row | * Field Labels Row: 1 | Import 0 rows of Data?: No, skip import | Record Name Field: WorkOrder ID

Fields 7 of 7 to import | Hide mapped fields

IMPORT FILE FIELD NAME	SALESFORCE FIELD NAME	SALESFORCE FIELD TYPE	ADD TO LAYOUTS	FIELD PREVIEW
✓ WorkOrder ID	WorkOrder ID	Text	✓	
✓ Email	Email	Email	✓	
✓ Service Type	Service Type	Picklist	✓	
✓ Description	Description	Text Area (Long)	✓	
✓ Location	Location	Picklist	✓	
✓ Priority	Priority	Picklist	✓	
✓ Status	Status	Picklist	✓	

Back | Next

ACTIVITY 3: Create Assignment Object

To create an object:

1. From the setup page --> Click on Object Manager --> Click on Create --> Click on Custom Object.

Setup | Home | Object Manager

Object Manager

51+ Items, Sorted by Label

Quick Find | Schema Builder | Create

Custom Object

Custom Object from Spreadsheet

2. Enter the label name --> Assignment
3. Plural label name --> Assignments

Field service workorder optimization

Custom Object Definition Edit Save Save & New Cancel

Custom Object Information

The singular and plural labels are used in tabs, page layouts, and reports.

Label Example: Account

Plural Label Example: Accounts

Starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name Example: Account

Description

Context-Sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window ☐ Open a window using a Visualforce page

Content Name

4. Enter Record Name Label and Format

- Record Name --> Assignment ID
- Data Type --> Auto Number
- Display Format --> A- {0000}
- Starting Number --> 1

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" a

Record Name Example: Account Name

Data Type Example: A-{0000} [What Is This?](#)

Display Format Example: A-{0000} [What Is This?](#)

Starting Number

5. Click on Allow reports, Allow search --> Save.

Field service workorder optimization

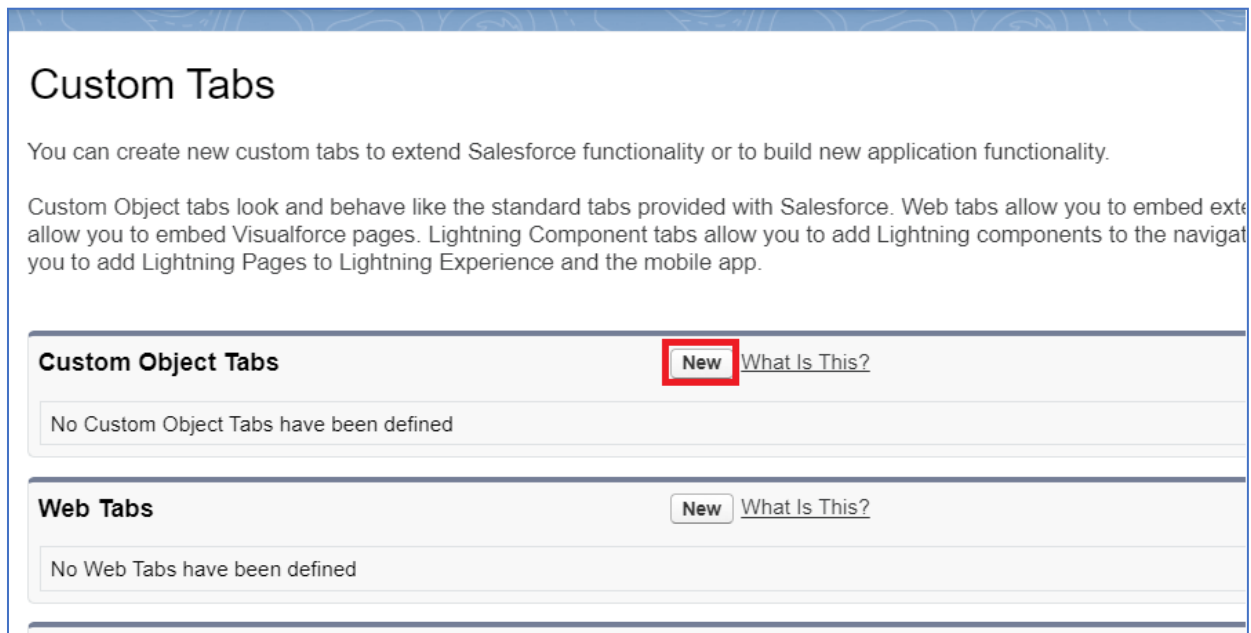
Milestone – 3

Tabs

Activity -1: Creating a Custom Tab

To create a Tab:(Assignment)

1. Go to setup page --> type Tabs in Quick Find bar --> click on tabs --> New (under custom object tab)



2. Select Object(Assignment) --> Select any tab style --> Next (Add to profiles page) keep it as default --> Next (Add to Custom App) keep it as default --> Save.

Field service workorder optimization

New Custom Object Tab

Step 1. Enter the Details

Choose the custom object for this new custom tab. Fill in other details.

Select an existing custom object or [create a new custom object now](#).

Object

Assignment

Tab Style

--None--

Assignment

(Optional) Choose a Home Page Custom Link to show as a splash page the first time your users click on this tab.

Splash Page Custom Link

--None--

Note: Tabs for Workorder & Technician objects do get created automatically. We do not need to create tabs for those objects.

Field service workorder optimization

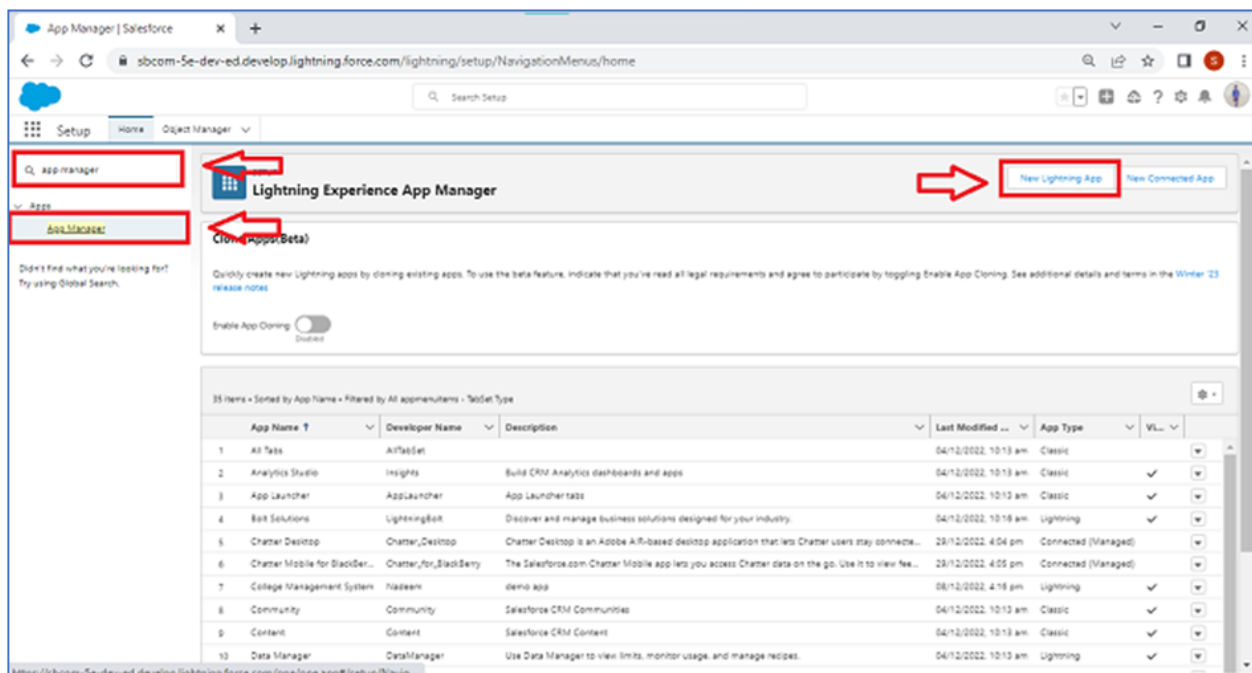
Milestone – 4

The Lightning App

Activity - 1: Create a Lightning App

To create a lightning app page:

1. Go to setup page --> search "app manager" in quick find --> select "app manager" --> click on new lightning App.



2. Fill the app name in app details and branding as follow
App Name: Field Service work Order Optimization
Developer Name: this will auto populated
Description: Give a meaningful description
Image: optional (if you want to give any image you can otherwise not mandatory)
Primary color hex value: keep this default
3. Then click Next --> (App option page) keep it as default --> Next --> (Utility Items) keep it as default --> Next.

Field service workorder optimization

New Lightning App

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

* App Name [?]
Name your app...

* Developer Name [?]
Enter a developer name...

Description [?]
Enter a description...

App Branding

Image [?]
Upload

Primary Color Hex Value
#0070C2

Org Theme Options
☐ Use the app's image and color instead of the org's custom theme

App Launcher Preview

Next

4. To Add Navigation Items:

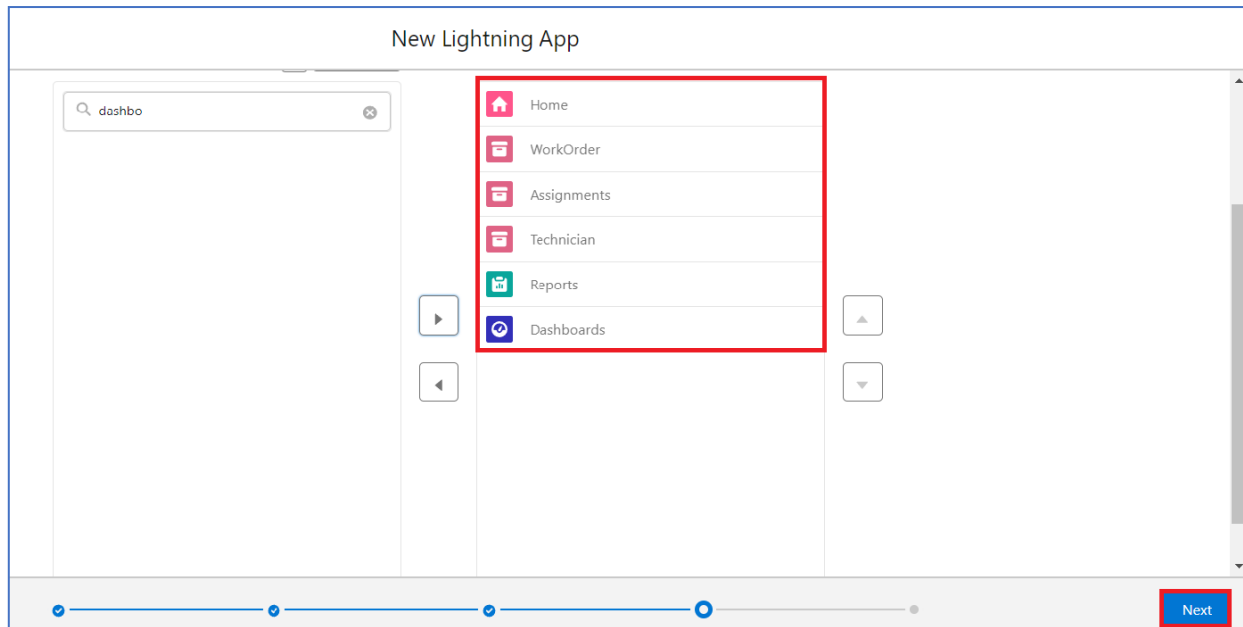
New Lightning App

dashbo

- Home
- WorkOrder
- Assignments
- Technician
- Reports
- Dashboards

Next

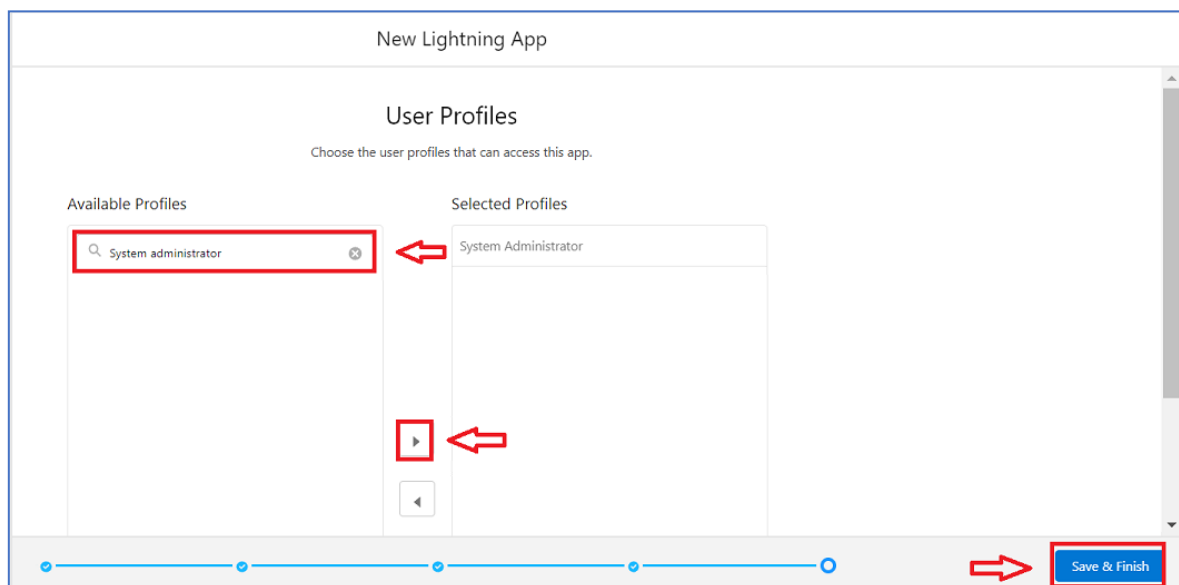
Field service workorder optimization



Search the items in the search bar (Home, work Order, Technician, Assignment, Reports, Dashboard) from the search bar and move it using the arrow button ? Next.

Note: select asset the custom object which we have created in the previous activity.

6. To Add User Profiles:



Search profiles (System administrator) in the search bar --> click on the arrow button --> save & finish.

Field service workorder optimization

Milestone – 5

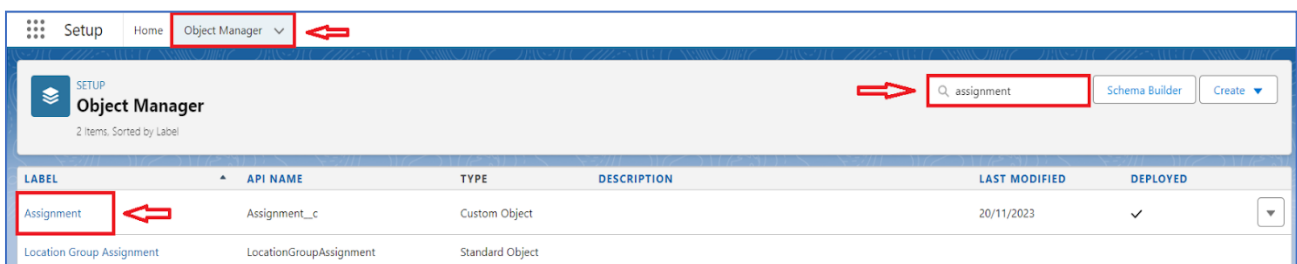
Fields & Relationship

Now it's time for you to think out of the box for your organization. You have successfully created the database objects for the organization but now all eyes turn on you as you have to define what sort of information the objects store which you have created. As a life saver of your organization, you come up with the idea of creating fields to store different types of data.

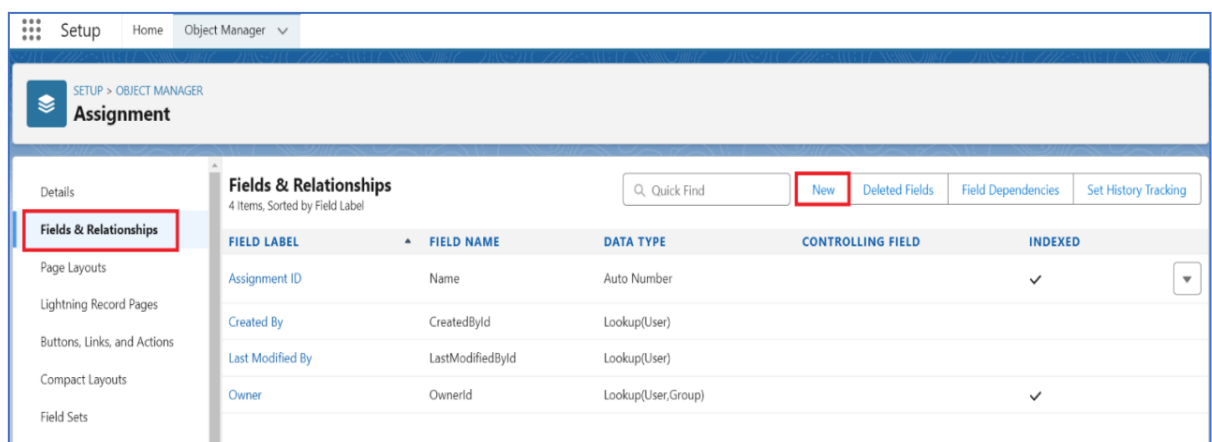
Activity - 1: Creating Lookup Field in Assignment Object

To create fields in an object:

1. Go to setup --> click on Object Manager --> type object name(Assignment) in quick find bar--> click on the object.



2. Now click on "Fields & Relationships" --> New



Field service workorder optimization

3. Select Data type as "Lookup".

Step 1. Choose the field type Step 1

[Next](#) [Cancel](#)

Specify the type of information that the custom field will contain.

Data Type

☐ None Selected Select one of the data types below.

☐ Auto Number A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.

☐ Formula A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.

☐ Roll-Up Summary A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.

☒ **Lookup Relationship** Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

☐ Master-Detail Relationship Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where:

- The relationship field is required on all detail records.

4. Click on Next

5. For field label related to: select "Work Order" object and click Next.

Note: Do not select other standard object with the same name for sake of ease copy the above and paste it.

Assignment Help for this Page

New Relationship

Step 2. Choose the related object Step 2

[Previous](#) [Next](#) [Cancel](#)

Select the other object to which this object is related.

Related To **WorkOrder**

[Previous](#) [Next](#) [Cancel](#)

7. Give Field Label as "Work Order ID" and click Next.

Assignment Help for this Page

New Relationship

Step 3. Enter the label and name for the lookup field Step 3 of 6

[Previous](#) [Next](#) [Cancel](#)

Field Label **WorkOrder ID**

Field Name **WorkOrder_ID**

Description

Help Text

8. Next --> Next --> Save & New.

Field service workorder optimization

Activity - 2: Manage your picklist values

1. From the setup page go to object manager
2. Search and Select Work Order object.
3. Go to fields & relationship, select Location field, scroll down to values and click "New".

Values					
New Reorder Replace Printable View Chart Colors Delete Selected Deactivate Selected Replace Selected					
☐ Action	Values	API Name	Default	Chart Colors	Modified By
☐ Edit Del Deactivate	Pune	Pune	☐	Assigned dynamically	demo project , 22/11/2023, 9:53 am
☐ Edit Del Deactivate	Hyderabad	Hyderabad	☐	Assigned dynamically	demo project , 22/11/2023, 9:53 am

4. Add the below values:
Nasik
Warangal
Nanded
5. Click Save.

Add Picklist Values

Location

Add one or more picklist values below. Each value should be on its own line and it is used for both a value's label and API name.

If a value matches an inactive value's API name, that value is reactivated with its previous label.

If a value matches an inactive value's label but not the API name, a new value is created.

Nasik
Warangal
Nanded

Save

Cancel

Page | 16

Field service workorder optimization

Activity - 3: Manage your picklist values

Add following values to the respective fields in Work Order object:

Field	Values
Priority	High
Service Type	Hardware repair Troubleshoot/Debugging Lane-Management

Activity - 4: Creating Formula Field in Work Order Object

1. Repeat step 1 and 2 mentioned in activity 1
2. Select Data type as "Formula" and click Next.
3. Give Field Label and Field Name as "Date" and select formula return type as "Date" and click next.

Step 2. Choose output type Step 2 of 5

Previous **Next** Cancel

Field Label Field Name

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity [i](#)

Formula Return Type

☐ None Selected Select one of the data types below.

☐ Checkbox Calculate a boolean value.
Example: `TODAY() > CloseDate`

☐ Currency Calculate a dollar or other currency amount and automatically format the field as a currency amount.
Example: `Gross Margin = Amount - Cost_c`

☒ Date Calculate a date, for example, by adding or subtracting days to other dates.
Example: `Reminder Date = CloseDate - 7`

☐ Date/Time Calculate a date/time, for example, by adding a number of hours or days to another date/time.
Example: `Next = NOW() + 1`

☐ Number Calculate a numeric value.
Example: `Fahrenheit = 1.8 * Celsius_c + 32`

☐ Percent Calculate a percent and automatically add the percent sign to the number.
Example: `Discount = (Amount - Discounted_Amount_c) / Amount`

☐ Text Create a text string, for example, by concatenating other text fields.
Example: `Full Name = LastName & ", " & FirstName`

☐ Time Calculate a time, for example, by adding a number of hours to another time.
Example: `Next = TIMEVALUE(NOW()) + 1`

Field service workorder optimization

4. Under Advanced Formula write down the formula and click "Check Syntax"

Formula: Created Date

Simple Formula Advanced Formula

Insert Field

Date (Date) =

CreatedDate

Check Syntax No syntax errors in merge fields or functions. (Compiled size: 20 characters)

5. Next--> Next--> Save.

Field service workorder optimization

Activity - 5: Creating Remaining fields for the respective objects

Now create the remaining fields using the data types mentioned in the table.

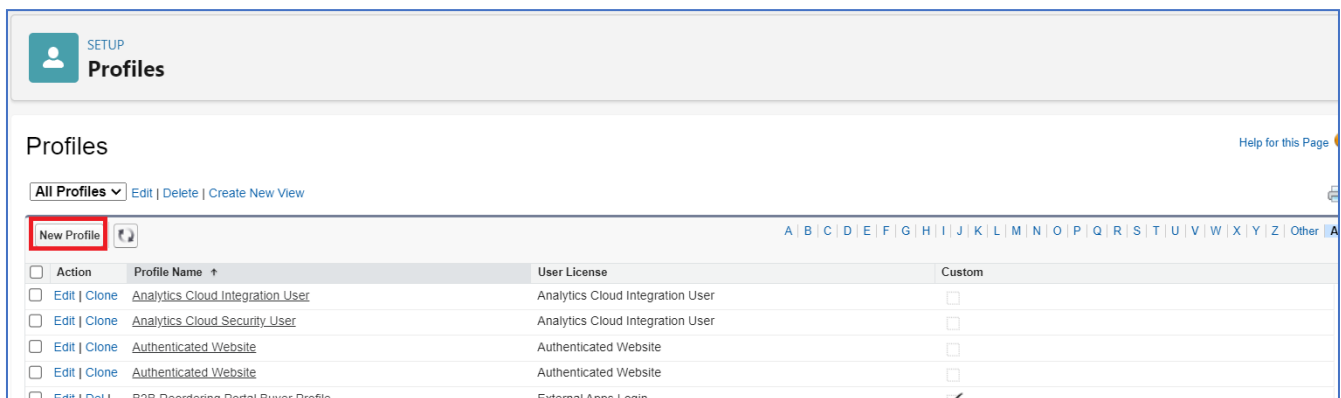
SNo	Object Name	Field	
1	Assignment		
		Field Name	Datatype
		• Technician ID	Lookup (Technician)
		• Assignment Date	Formula: return type: Date (WorkOrder_ID__r. Date__c)
		• Completion Date	Formula: return type: Date IF (ISPICKVAL (WorkOrder_ID__r. Status__c , 'Resolved'), WorkOrder_ID__r.LastModifiedDate , NULL)

Field service workorder optimization

Milestone - 6 Profiles

Activity - 1: Technician Profile

1. Go to setup --> type profiles in quick find box --> click on profiles --> click on new profile.



SETUP
Profiles

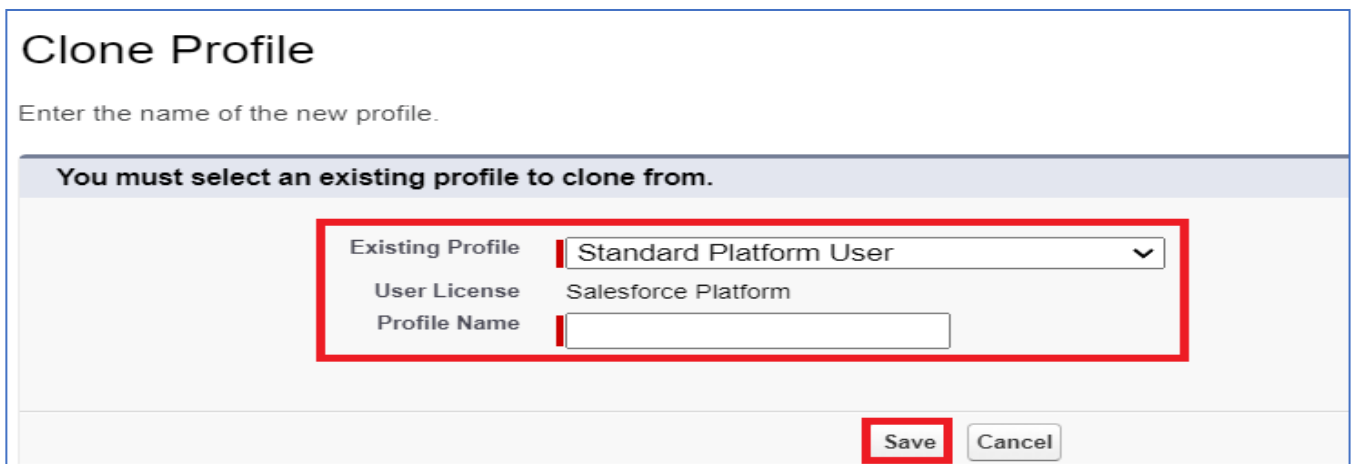
Profiles [Help for this Page](#)

[All Profiles](#) [Edit](#) [Delete](#) [Create New View](#)

[New Profile](#) [A](#) [B](#) [C](#) [D](#) [E](#) [F](#) [G](#) [H](#) [I](#) [J](#) [K](#) [L](#) [M](#) [N](#) [O](#) [P](#) [Q](#) [R](#) [S](#) [T](#) [U](#) [V](#) [W](#) [X](#) [Y](#) [Z](#) [Other](#) [A](#)

Action	Profile Name	User License	Custom
Edit Clone	Analytics Cloud Integration User	Analytics Cloud Integration User	☐
Edit Clone	Analytics Cloud Security User	Analytics Cloud Integration User	☐
Edit Clone	Authenticated Website	Authenticated Website	☐
Edit Clone	Authenticated Website	Authenticated Website	☐
Edit Delete	R2B Reordering Portal Buyer Profile	External Apps Login	☒

2. Select 'Standard Platform User' for existing profile and give 'Technician' for Profile Name and click on Save.



Clone Profile

Enter the name of the new profile.

You must select an existing profile to clone from.

Existing Profile:

User License: Salesforce Platform

Profile Name:

[Save](#) [Cancel](#)

3. While still on the profile page, then click Edit.
4. Scroll down to Custom Object Permissions and Give Read only access permissions for Technician, Work Order and Assignment objects and field access permission as shown below:

Field service workorder optimization

Custom Object Permissions						
	Basic Access				Data Administration	
	Read	Create	Edit	Delete	View All 1	Modify All 1
Assets	☐	☐	☐	☐	☐	☐
Asset Services	☐	☐	☐	☐	☐	☐
Assignments	☒	☐	☐	☐	☐	☐
Billings	☐	☐	☐	☐	☐	☐
Bookings	☐	☐	☐	☐	☐	☐
Candidates	☐	☐	☐	☐	☐	☐
Child object	☐	☐	☐	☐	☐	☐
Crews	☐	☐	☐	☐	☐	☐
Customer Orders	☐	☐	☐	☐	☐	☐
Employees	☐	☐	☐	☐	☐	☐
Employment Websites	☐	☐	☐	☐	☐	☐
Flights	☐	☐	☐	☐	☐	☐
Items	☐	☐	☐	☐	☐	☐
Jewel Customers	☐	☐	☐	☐	☐	☐
Job Applications	☐	☐	☐	☐	☐	☐
Lead Scoring Rules	☐	☐	☐	☐	☐	☐
Leaves	☐	☐	☐	☐	☐	☐
Parent object 1	☐	☐	☐	☐	☐	☐
Parent object 2	☐	☐	☐	☐	☐	☐
Passengers	☐	☐	☐	☐	☐	☐
Positions	☐	☐	☐	☐	☐	☐
Prices	☐	☐	☐	☐	☐	☐
Projects	☐	☐	☐	☐	☐	☐
ProjectTasks	☐	☐	☐	☐	☐	☐
Reviews	☐	☐	☐	☐	☐	☐
Sessions	☐	☐	☐	☐	☐	☐
Students	☐	☐	☐	☐	☐	☐
StudentSessions	☐	☐	☐	☐	☐	☐
Technician	☒	☐	☐	☐	☐	☐
Trainers	☐	☐	☐	☐	☐	☐
WorkOrder	☒	☐	☐	☐	☐	☐

5. Scroll down and Click on Save.
6. Now from the profile detail page scroll down to custom field level security click on view next to Work Order object.

Custom Field-Level Security	
Asset	View
Asset Service	View
Assignment	View
Billing	View
Booking	View
Candidate	View
Child object	View
Crew	View
Customer Order	View
Employee	View
Employment Website	View
Flight	View
Item	View
Jewel Customer	View
Job Application	View
Knowledge	View
Lead Scoring Rule	View
Leave	View
Parent object 1	View
Parent object 2	View
Passenger	View
Position	View
Price	View
Project	View
ProjectTask	View
Review	View
Sessions	View
Student	View
StudentSession	View
Technician	View
Trainer	View
WorkOrder	View

7. Click on Save.

Field service workorder optimization

Milestone – 7

Users

Activity - 1: Create User

1. Go to setup --> type users in quick find box --> select users --> click New user.
2. Fill in the fields
 - First Name: Elina
 - Last Name: Gilbert
 - Alias: Give a Alias Name
 - Email id: Give your Personal Email id
 - Username: Username should be in this form: text@text.text
 - Nick Name: Give a Nickname
 - Role:
 - User license: Salesforce Platform
 - Profiles: Technician

New User

Help for this Page

User Edit Save Save & New Cancel

General Information

First Name Elina

Last Name Gilbert

Alias elina

Email nadeem@thesmartbridge.coi

Username elina@smart.com

Nickname elina

Title

Company

Department

Division

Role <None Specified>

User License Salesforce Platform

Profile Technician

Active ☒

Marketing User ☐

Offline User ☐

Knowledge User ☐

Flow User ☐

Service Cloud User ☐

Site.com Contributor User ☐

Site.com Publisher User ☐

WDC User ☐

= Required Information

3. Save

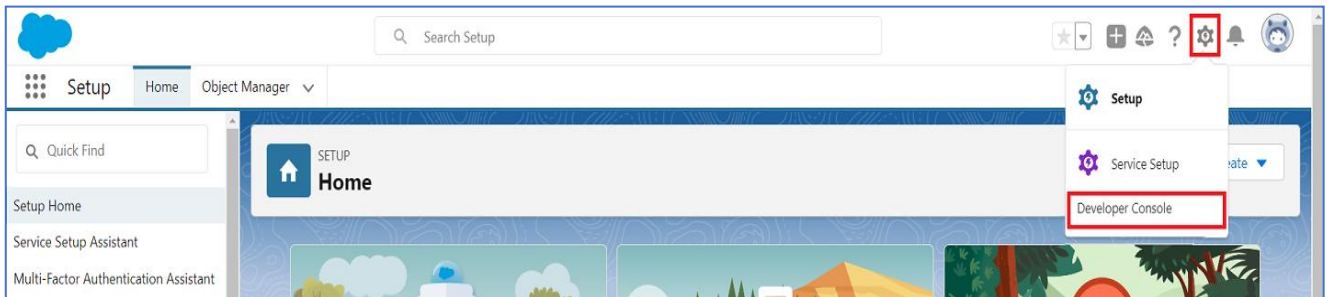
Field service workorder optimization

Milestone – 8

Apex Trigger

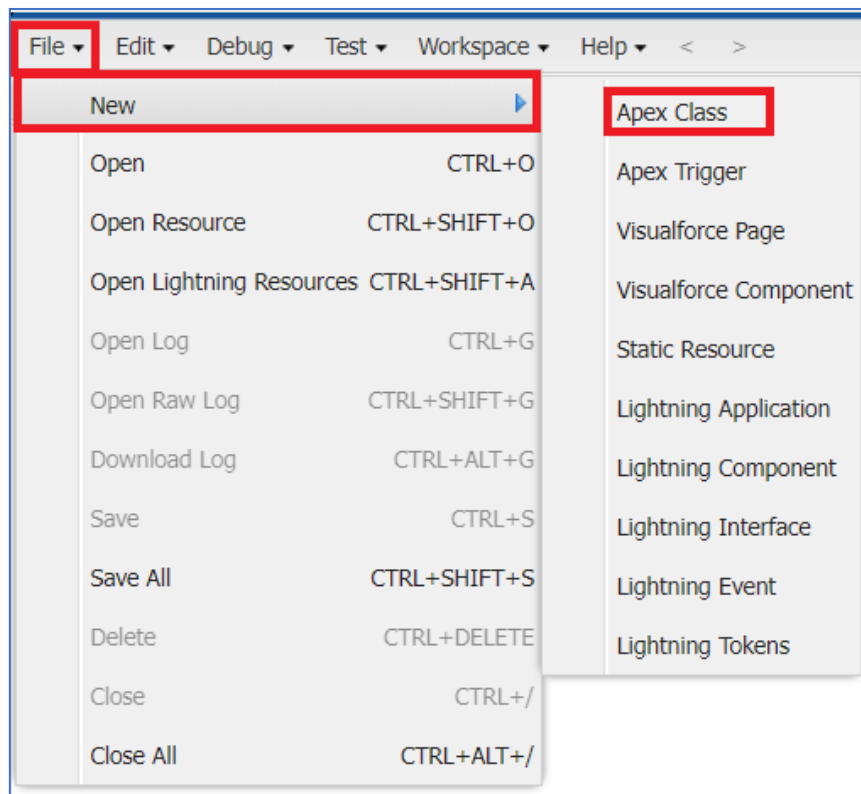
Activity - 1: Create an Apex Class

1. Go to Setup --> Click on the gear icon --> Select Developer Console.



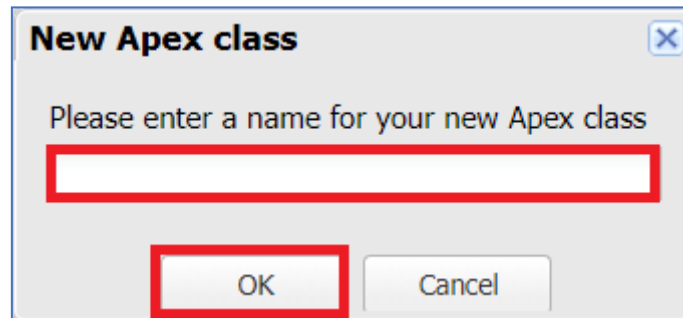
2. Then we can see the Developer console. Click on the developer console and you will navigate to a new console window.

3. To create a new Apex Class follow the below steps:
Click on the file --> New --> Apex Class.



Field service workorder optimization

4. Give the Apex Class name as "WorkOrderClass".



5. Click ok.

6. Now write the code logic here

```
1 public class WorkOrderClass {
2     public static void workOrder(List<WorkOrder__c> newListWorkOrder){
3         Map<Integer, List<String>> maptotech = new Map<Integer, List<String>>();
4         Integer num = 0;
5         List<WorkOrder__c> properWo = new List<WorkOrder__c>();
6         List<Assignment__c> lstAssignment = new List<Assignment__c>();
7         List<Technician__c> technicianToAssignment = new List<Technician__c>();
8         for(WorkOrder__c iter : newListWorkOrder){
9             List<String> lststring = new List<String>();
10            If(iter.Service_Type__c != null && iter.Location__c != null ){
11                num = num+1;
12                properWo.add(iter);
13                lststring.add(iter.Service_Type__c);
14                lststring.add(iter.Location__c);
15            }
16            maptotech.put(num, lststring);
17        }
18    }
19    Map<Integer, Id> techId = new Map<Integer, Id>();
20    Map<Id, Technician__c> allTechnician = new Map<Id, Technician__c>([SELECT Id, Name, Phone__c, Location__c, Skills__c, Availability__c, Name__c, Email__c FROM Technician__c]);
21    Integer num2 = 0;
22    For(Technician__c T : allTechnician.values()){
23        num2 = num2+1;
24        if(maptotech.get(num2) != null){
25            List<String> valofmap = maptotech.get(num2);
26            system.debug('error 1 ----> the maptotech is empty ----> ' + maptotech.get(num2));
27            if(valofmap.contains(t.Skills__c) && ValofMap.contains(t.Location__c) && t.Availability__c == 'Available'){
28                techid.put(num2, t.Id);
29            }
30        }
31    }
32 }
33 Integer num3 = 0;
34 For(WorkOrder__c W : properWo){
```

Field service workorder optimization

Source Code:

```
public class WorkOrderClass {
    public static void workOrder(List<WorkOrder__C> newListWorkOrder){
        Map<Integer, List<String>> maptotech = new map<Integer,List<String>>();
        integer num = 0;
        List<WorkOrder__c> properWo = new List<WorkOrder__c>();
        List<Assignment__c> lstAssignment = new List<Assignment__c>();
        List<Technician__c> technicianToAssignment = new List<Technician__c>();
        for(WorkOrder__c iter : newListWorkOrder){
            List<String> lststring = new List<string>();
            If(iter.Service_Type__c != null && iter.Location__c != null ){
                num = num+1;
                properWo.add(iter);
                lststring.add(iter.Service_Type__c);
                lststring.add(iter.Location__c);

                maptotech.put(num,lststring);
            }
        }
        Map<integer,Id> techId = new Map<integer,Id>();
        Map<Id,Technician__c> allTechnician = new Map<Id,Technician__c>([SELECT Id, Name,
        Phone__c, Location__c, Skills__c, Availability__c, Name__c, Email__c FROM Technician__c]);
        integer num2 = 0;
        For(Technician__c T : allTechnician.values()){
            num2 = num2+1;
            if(maptotech.get(num2) != null){
                List<string> valofmap = maptotech.get(num2);
                system.debug('error 1 ----> the maptotech is empty ---> ' + maptotech.get(num2));
                if(valofMap.contains(t.Skills__c) && ValofMap.contains(t.Location__c) && t.Availability__c ==
                'Available'){
                    techid.put(num2,t.Id);
                }
            }
        }
        integer num3 = 0;
```

Field service workorder optimization

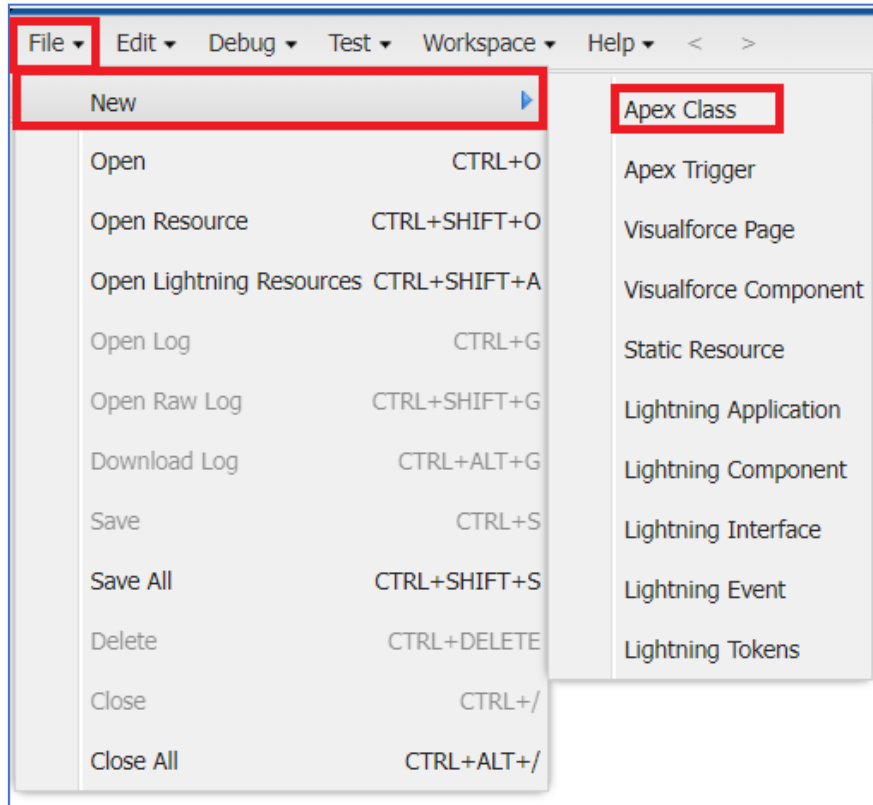
```
For(WorkOrder__c W: properWo){
    num3 = num3 + 1;
    Assignment__c A = new Assignment__c ();
    A.WorkOrder_ID__c = W.Id;
    A.Technician_ID__c = techid.get(num3);
    lstAssignment.add(A);
}
If(!lstAssignment.IsEmpty()){
    insert lstAssignment;
}
}
```

7. Save the code. (click on file --> Save)

Field service workorder optimization

Activity - 2 : Create an Apex Trigger

1. To create a new Apex Class follow the below steps:
Click on the file --> New --> Apex Class.

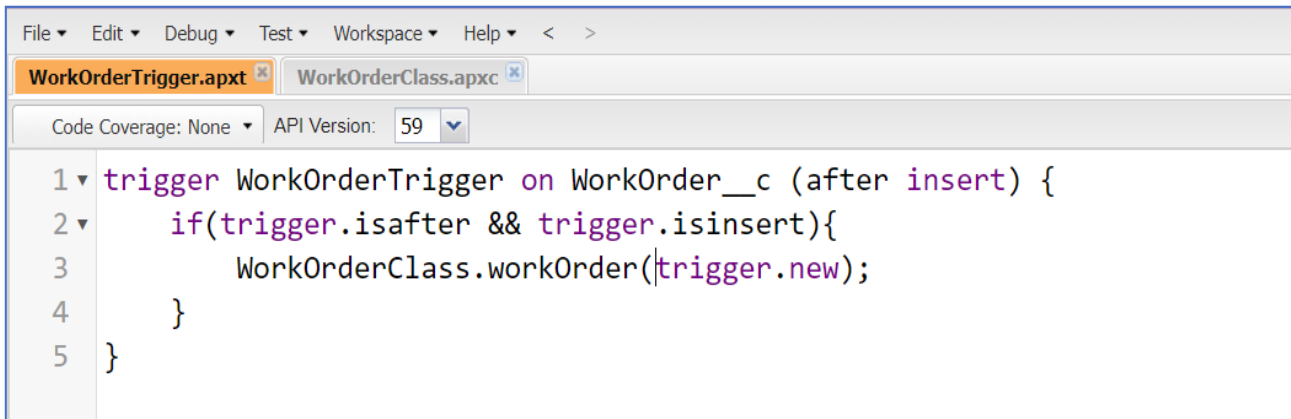


2. Give the Apex Trigger name as "WorkOrderTrigger", and select "WorkOrder__c" from the dropdown for sObject.

A screenshot of the 'New Apex Trigger' dialog box in the Salesforce IDE. The dialog has a title bar with the text 'New Apex Trigger' and a close button. Inside, there are two labels: 'Name:' and 'sObject:'. Next to 'Name:' is a text input field. Next to 'sObject:' is a dropdown menu. Both the input field and the dropdown menu are highlighted with red rectangular boxes. At the bottom right of the dialog, there is a 'Submit' button, which is also highlighted with a red rectangular box.

Field service workorder optimization

3. Click Submit.
4. Now write the code logic here



```
1 trigger WorkOrderTrigger on WorkOrder__c (after insert) {
2     if(trigger.isafter && trigger.isinsert){
3         WorkOrderClass.workOrder(trigger.new);
4     }
5 }
```

Source Code:

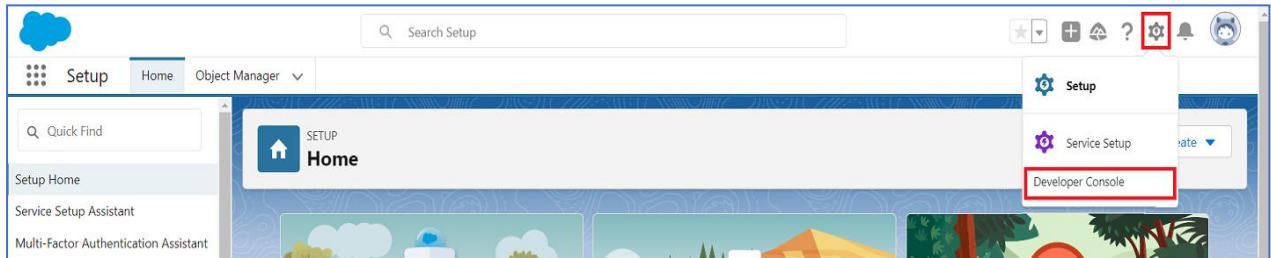
```
trigger WorkOrderTrigger on WorkOrder__c (after insert) {
    if(trigger.isafter && trigger.isinsert){
        WorkOrderClass.workOrder(trigger.new);
    }
}
```

5. Save the code. (click on file --> Save).

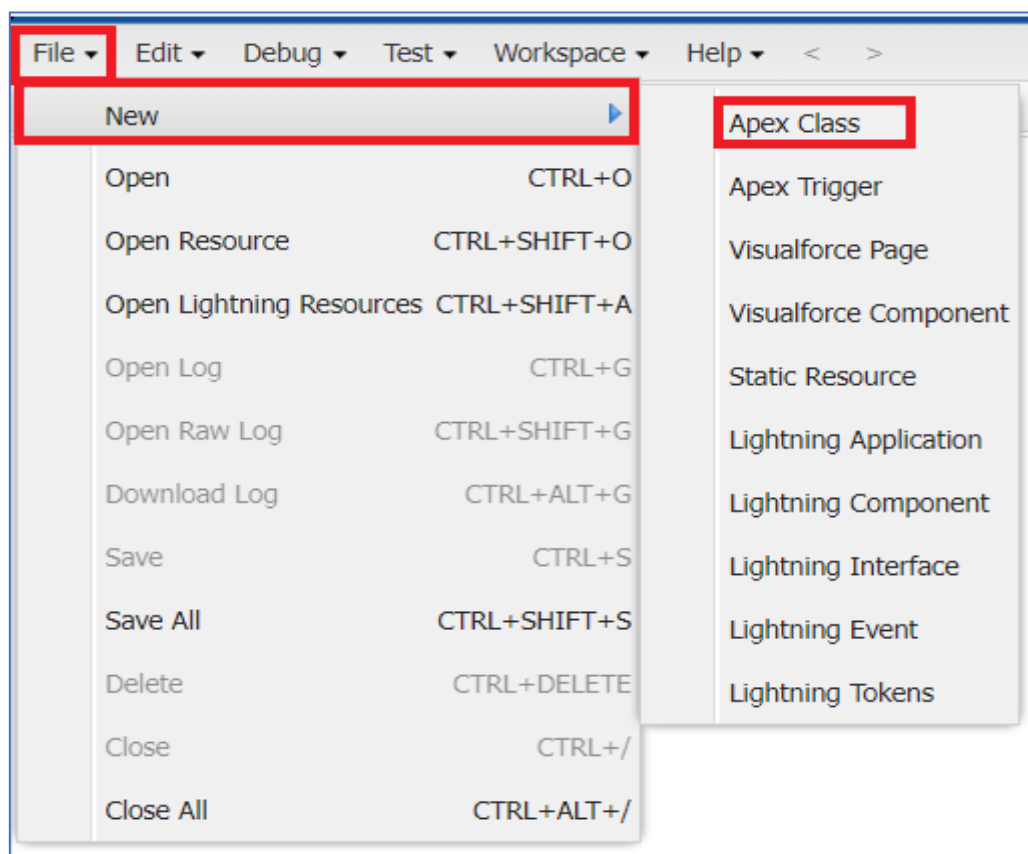
Field service workorder optimization

Activity - 3: Create an Apex Class

1. Go to Setup --> Click on the gear icon --> Select Developer Console.

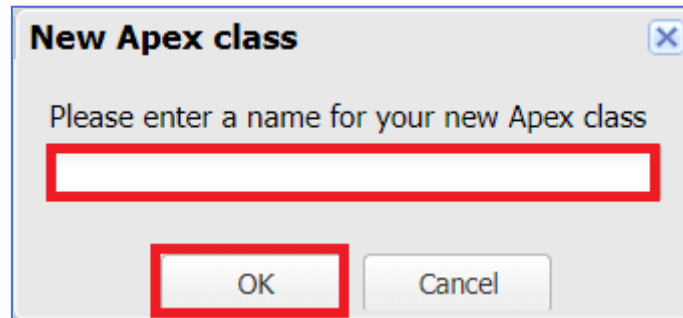


2. Then we can see the Developer console. Click on the developer console and you will navigate to a new console window.
3. To create a new Apex Class follow the below steps:
Click on the file --> New --> Apex Class.



4. Give the Apex Class name as "AssigningEmail".

Field service workorder optimization



- Click ok.
- Now write the code logic here



Source Code:

```
public class AssigningEmail {
    public static void sendEmailmsg(List<Assignment__c> assRec){
        List<messaging.SingleEmailMessage> myVar = new List<messaging.SingleEmailMessage>();
        Map<id,Technician__c> technicians = new Map<id,Technician__c>([SELECT Id, Phone__c,
        Location__c, Skills__c, Name__c, Email__c, Availability__c, Name FROM Technician__c]);
        try{
            for(Assignment__c con : assRec){
                if(con.Technician_ID__c != null){
                    messaging.SingleEmailMessage mail = new messaging.SingleEmailMessage();
                    List<String> sendTo = new List<String>();
                    sendTo.add(technicians.Get(con.Technician_ID__c).Email__c);
                    mail.setToAddresses(sendTo);
                    string subject = 'WorkOrder Assignment ';
```

Field service workorder optimization

```
mail.setSubject(subject);
string body = 'The following WorkOrder has been assigned to you ';
mail.setHTMLbody(body);
myVar.add(mail);
    }
}
Messaging.sendEmail(myvar);
}
catch(exception e){
    system.debug('Error -----> ' + e.getMessage());
}
}
```

Source Code:

```
public class AssigningEmail {
    public static void sendEmailmsg(List<Assignment__c> assRec){
        List<messaging.SingleEmailMessage> myVar = new List<messaging.SingleEmailMessage>();
        Map<id,Technician__c> technicians = new Map<id,Technician__c>([SELECT Id, Phone__c,
Location__c, Skills__c, Name__c, Email__c, Availability__c, Name FROM Technician__c]);
        try{
            for(Assignment__c con : assRec){
                if(con.Technician_ID__c != null){
                    messaging.SingleEmailMessage mail = new messaging.SingleEmailMessage();
                    List<String> sendTo = new List<String>();
                    sendTo.add(technicians.Get(con.Technician_ID__c).Email__c);
                    mail.setToAddresses(sendTo);
                    string subject = 'WorkOrder Assignment ';
                    mail.setSubject(subject);
                    string body = 'The following WorkOrder has been assigned to you ';
                    mail.setHTMLbody(body);
                    myVar.add(mail);
                }
            }
            Messaging.sendEmail(myvar);
        }
        catch(exception e){
            system.debug('Error -----> ' + e.getMessage());
        }
    }
}
```

Field service workorder optimization

```
}  
}  
}
```

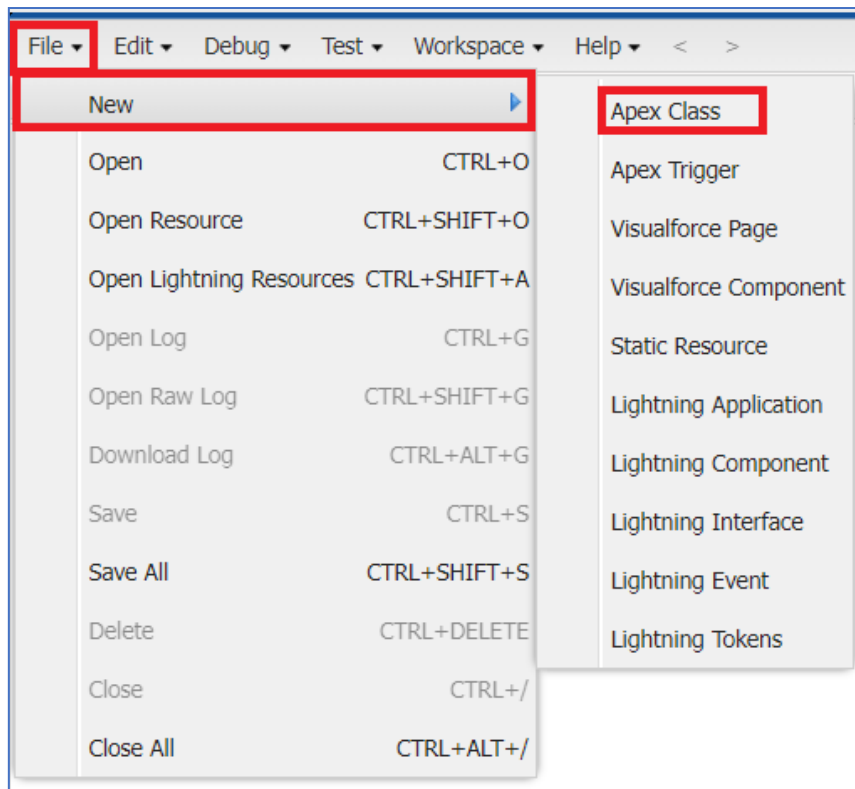
6. Save the code. (click on file --> Save)

Field service workorder optimization

Activity - 4: Create an Apex Trigger

To create a new Apex Class, follow the below steps:

1. Click on the file --> New --> Apex Class.

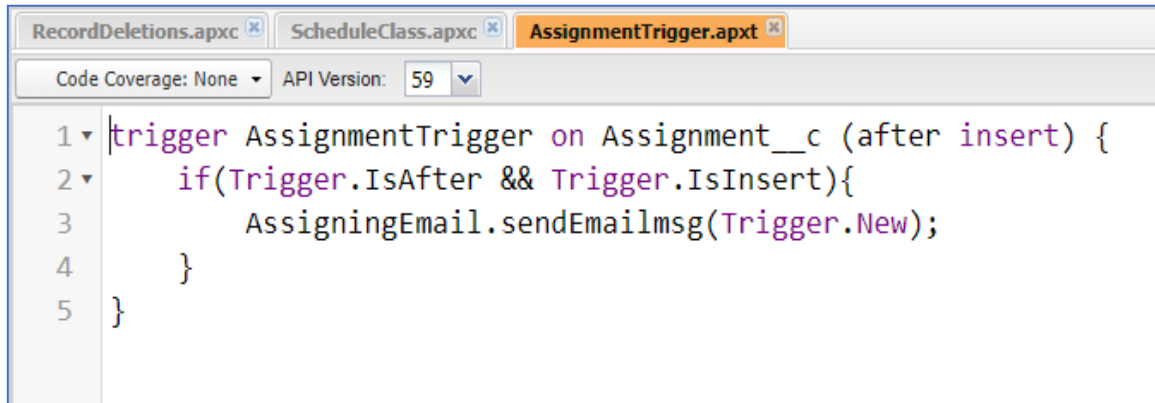


2. Give the Apex Trigger name as "AssignmentTrigger", and select "Assignment__c" from the dropdown for sObject.

A screenshot of the 'New Apex Trigger' dialog box in Salesforce. It has a title bar with a close button. Inside, there are two fields: 'Name:' followed by a text input box, and 'sObject:' followed by a dropdown menu. Both the input box and the dropdown menu are highlighted with red boxes. At the bottom right of the dialog, there is a 'Submit' button, which is also highlighted with a red box.

Field service workorder optimization

3. Click Submit.
4. Now write the code logic here



```
RecordDeletions.apxc | ScheduleClass.apxc | AssignmentTrigger.apxt
Code Coverage: None | API Version: 59
1 trigger AssignmentTrigger on Assignment__c (after insert) {
2     if (Trigger.IsAfter && Trigger.IsInsert){
3         AssigningEmail.sendEmailmsg(Trigger.New);
4     }
5 }
```

Source Code:

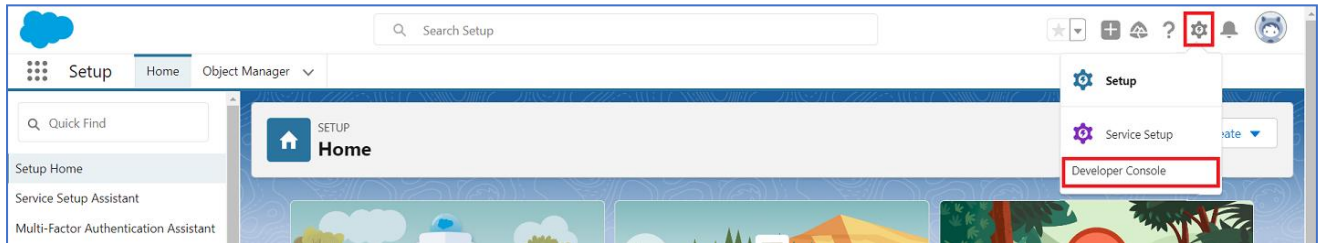
```
trigger AssignmentTrigger on Assignment__c (after insert) {
    if (Trigger.IsAfter && Trigger.IsInsert){
        AssigningEmail.sendEmailmsg(Trigger.New);
    }
}
```

5. Save the code.(click on file --> Save)

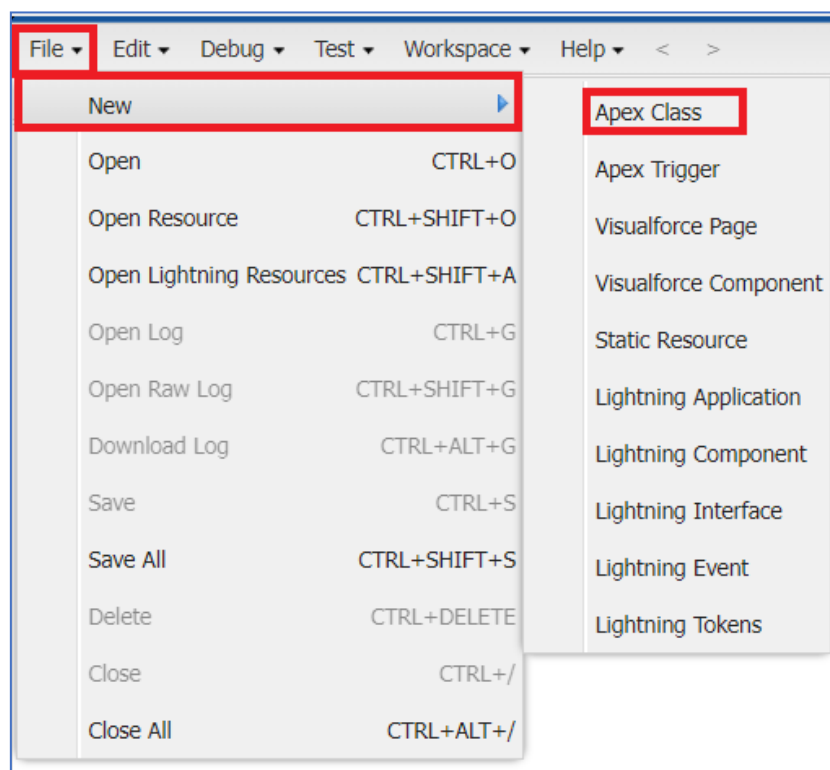
Field service workorder optimization

Activity - 5: Create an Apex Class

1. Go to Setup --> Click on the gear icon --> Select Developer Console.

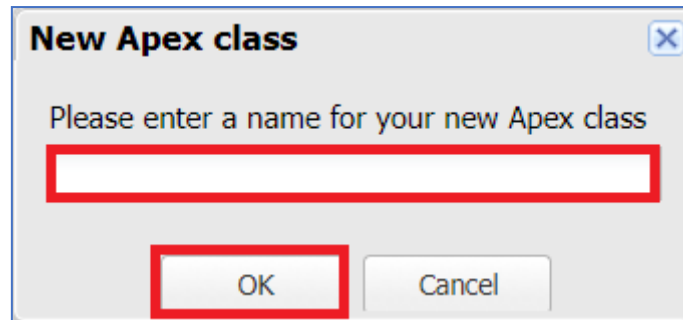


2. Then we can see the Developer console. Click on the developer console and you will navigate to a new console window.
3. To create a new Apex Class follow the below steps:
Click on the file --> New --> Apex Class.



4. Give the Apex Class name as "CompletionMail".

Field service workorder optimization



5. Click ok.

6. Now write the code logic here

```
RecordDeletions.apxc | ScheduleClass.apxc | CompletionMail.apxc
Code Coverage: None | API Version: 59
1 public class CompletionMail {
2     public static void sendEmailMsg(List<WorkOrder__c> workOrderList){
3         List<messaging.SingleEmailMessage> myVar = new List<messaging.SingleEmailMessage>();
4         for(WorkOrder__c con : workOrderList){
5             if(con.Status__c == 'Resolved'){
6                 messaging.SingleEmailMessage mail = new messaging.SingleEmailMessage();
7                 List<String> sendTo = new List<String>();
8                 sendTo.add(con.Email__c);
9                 mail.setToAddresses(sendTo);
10                string subject = 'Status Updated';
11                mail.setSubject(subject);
12                string body = 'email body ';
13                mail.setHTMLbody(body);
14                myVar.add(mail);
15            }
16        }
17        Messaging.sendEmail(myvar);
18    }
19 }
```

Field service workorder optimization

Source Code:

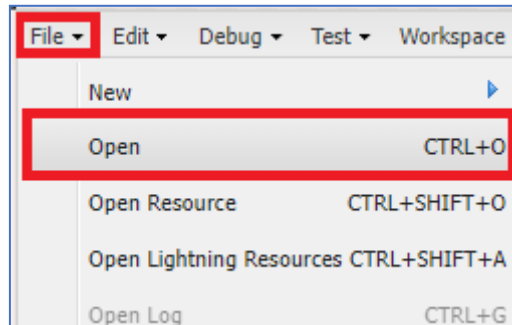
```
public class CompletionMail {
    public static void sendEmailMsg(List<WorkOrder__c> workOrderList){
        List<messaging.SingleEmailMessage> myVar = new List<messaging.SingleEmailMessage>();
        for(WorkOrder__c con : workOrderList){
            if(con.Status__c == 'Resolved'){
                messaging.SingleEmailMessage mail = new messaging.SingleEmailMessage();
                List<String> sendTo = new List<String>();
                sendTo.add(con.Email__c);
                mail.setToAddresses(sendTo);
                string subject = 'Status Updated';
                mail.setSubject(subject);
                string body = 'email body ';
                mail.setHTMLbody(body);
                myVar.add(mail);
            }
        }
        Messaging.sendEmail(myvar);
    }
}
```

7. Save the code. (click on file --> Save)

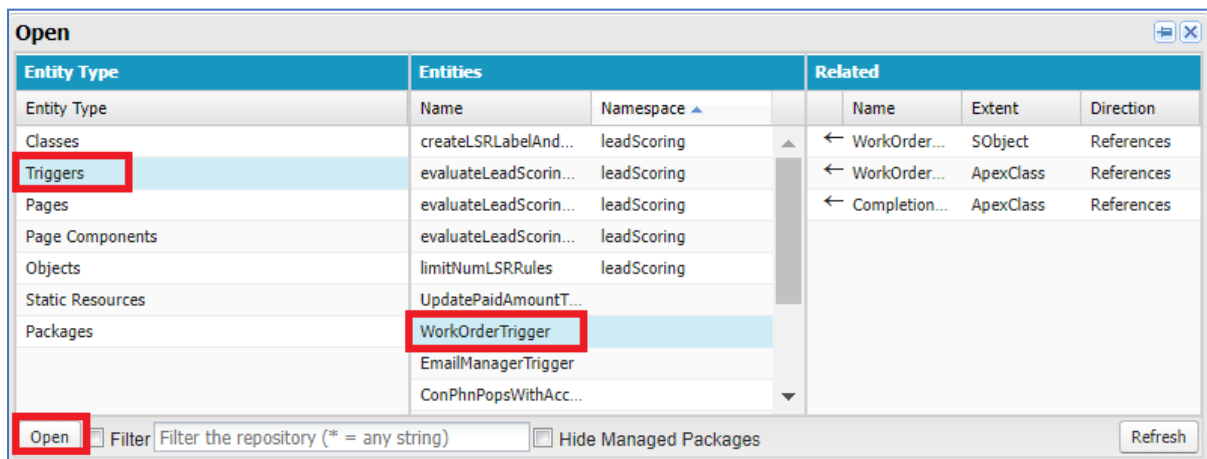
Field service workorder optimization

Activity - 6 : Create an Apex Trigger

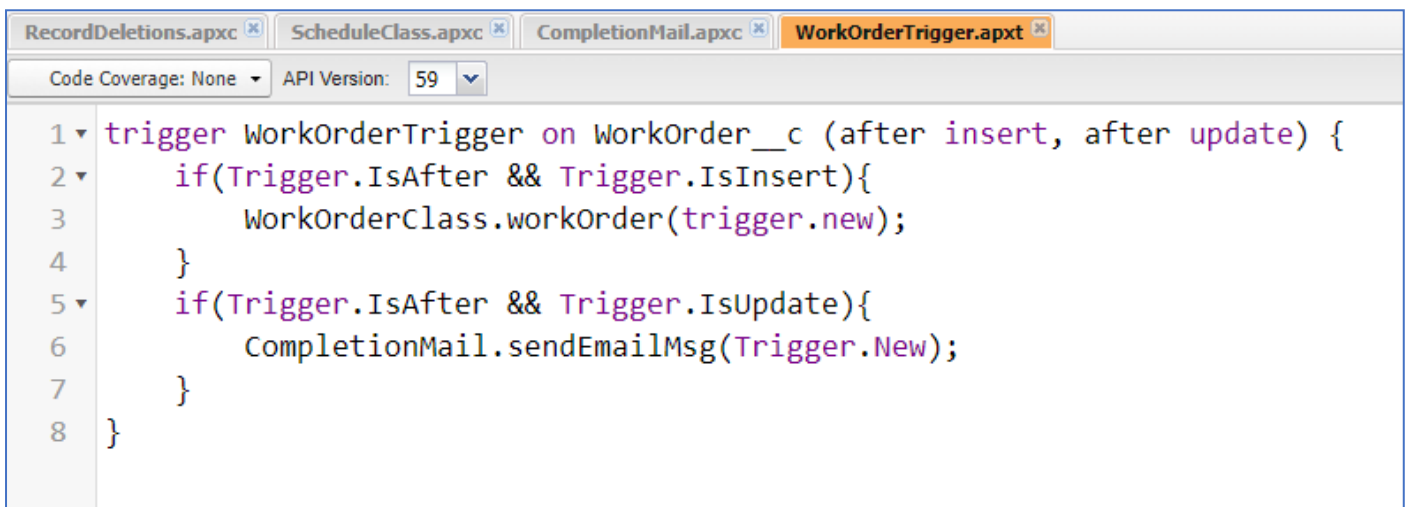
1. Click on the file --> Open.



2. A pop-up window opens click on Triggers, then select "WorkOrderTrigger" and click on "Open"



3. Now write the code logic here.



Field service workorder optimization

Source Code:

```
WorkOrderClass.workOrder(trigger.new);  
}  
if(trigger.IsAfter && Trigger.IsUpdate){  
    CompletionMail.sendEmailMsg(trigger.New);  
}  
}
```

5. Save the code.(click on file --> Save)

Field service workorder optimization

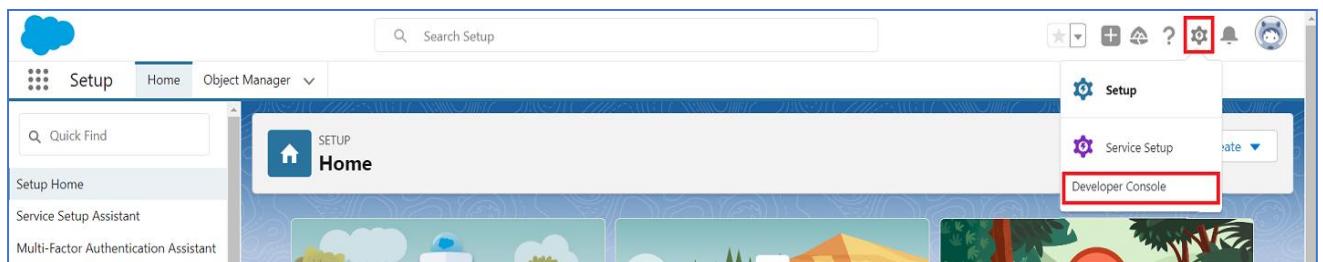
Activity - 7: Create an Asynchronous Apex Class

Create an Apex Class to Delete all the WorkOrder records which meets the following criteria

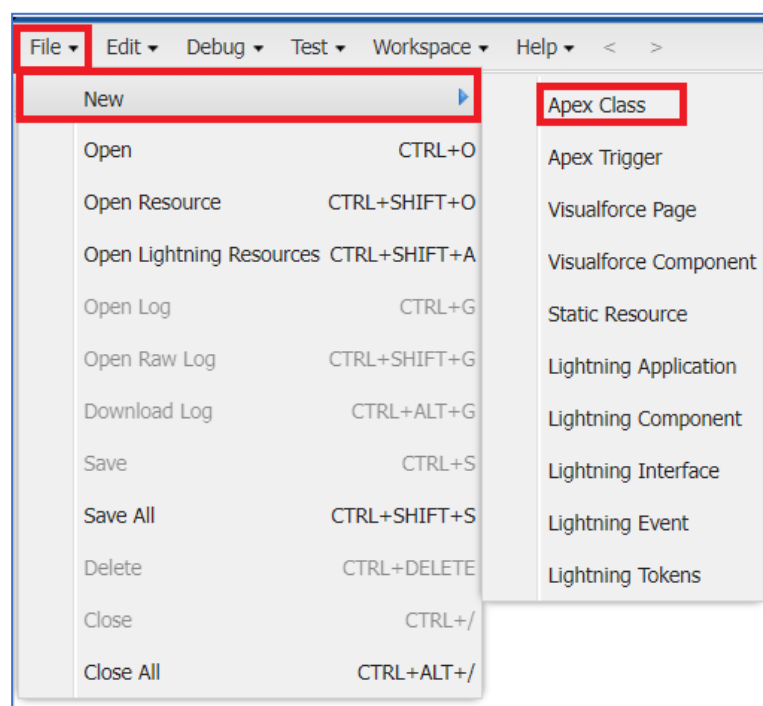
1. Completed date should be more than 30 days.
2. Status should be 'Resolved'.

Create an Apex Class

1. Go to Setup --> Click on the gear icon --> Select Developer Console.

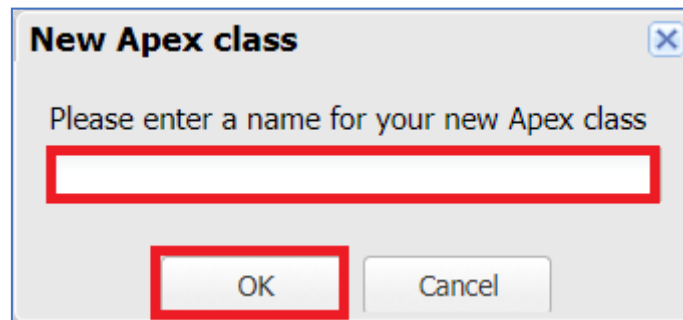


2. Then we can see the Developer console. Click on the developer console and you will navigate to a new console window.
3. To create a new Apex Class, follow the below steps:
Click on the file --> New --> Apex Class.

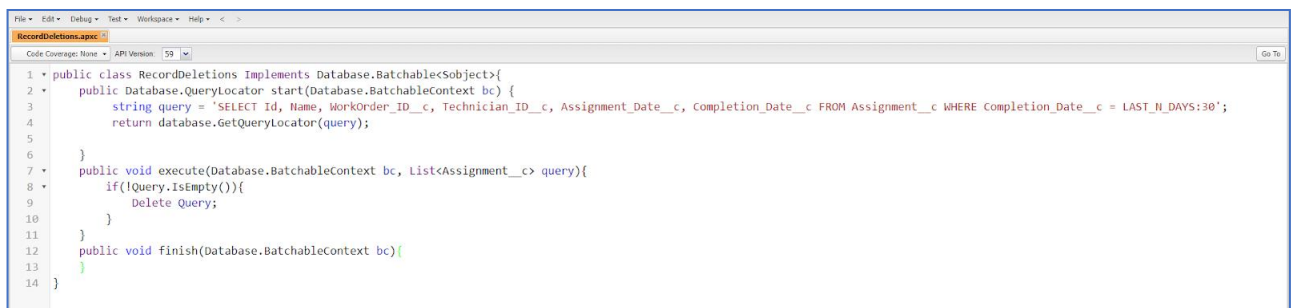


Field service workorder optimization

4. Give the Apex Class name as "RecordDeletion".



5. Click ok.
6. Now write the code logic here



Sample code:

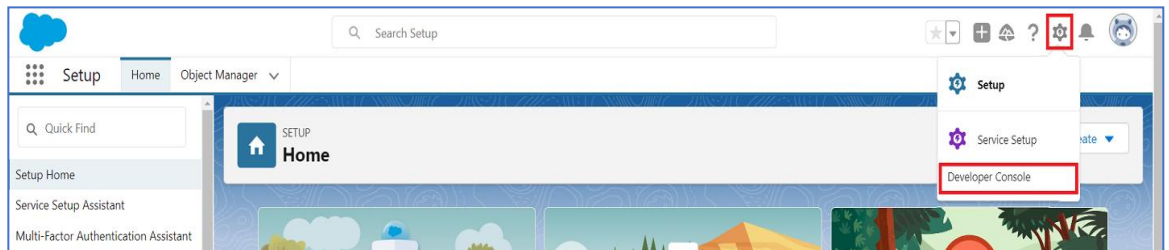
```
public class RecordDeletions implements Database.Batchable<Sobject>{
public Database.QueryLocator start(Database.BatchableContext bc) {
    string query = 'SELECT Id, Name, WorkOrder_ID__c, Technician_ID__c,
Assignment_Date__c, Completion_Date__c FROM Assignment__c WHERE
Completion_Date__c = LAST_N_DAYS:30';
    return database.GetQueryLocator(query);
}
public void execute(Database.BatchableContext bc, List<Assignment__c> query){
    if(!Query.IsEmpty()){
        i. Delete Query;
    }
}
public void finish(Database.BatchableContext bc){
}
}
```

7. Save the code.(click on file --> Save)

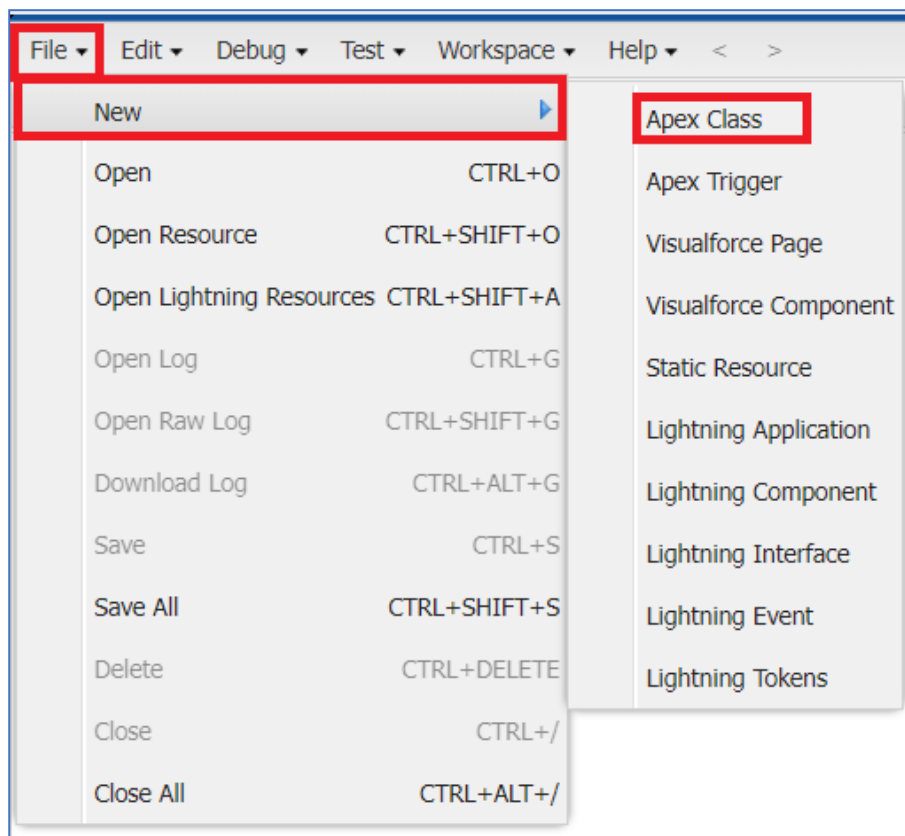
Field service workorder optimization

ACTIVITY 8 : Create an Apex Schedule Class

1. Go to Setup --> Click on the gear icon --> Select Developer Console.

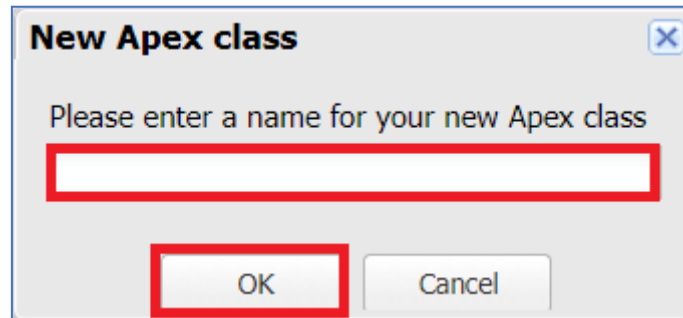


2. Then we can see the Developer console. Click on the developer console and you will navigate to a new console window.
3. To create a new Apex Class follow the below steps:
Click on the file --> New --> Apex Class.

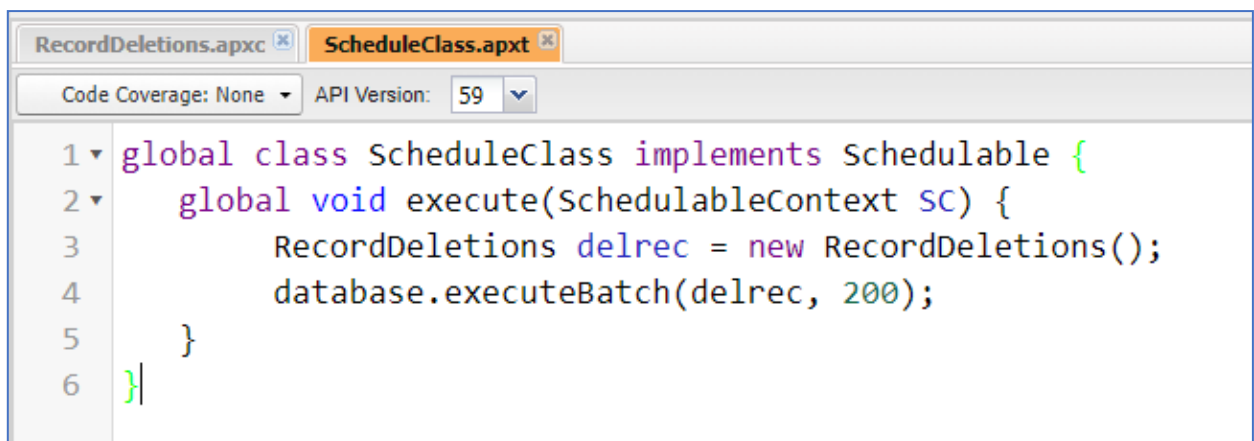


4. Give the Apex Class name as "ScheduleClass".

Field service workorder optimization



5. Click ok.
6. Now write the code logic here



Source Code:

```
global class ScheduleClass implements Schedulable {  
    global void execute(SchedulableContext SC) {  
        RecordDeletions delrec = new RecordDeletions();  
        database.executeBatch(delrec, 200);  
    }  
}
```

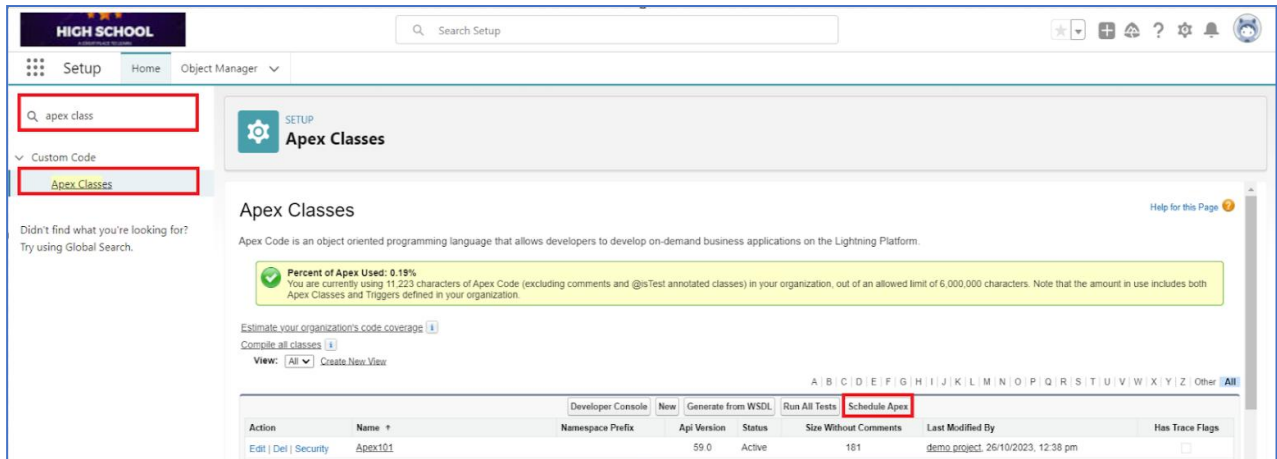
7. Save the code.(click on file ? Save)

Field service workorder optimization

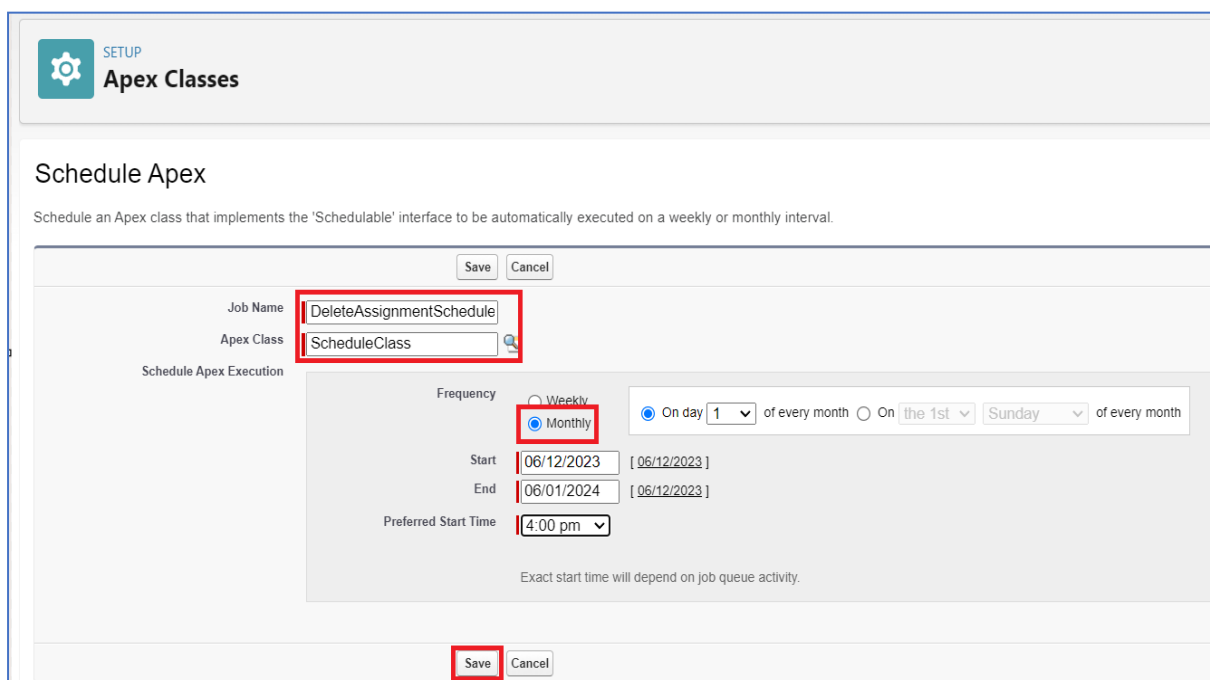
Activity - 9: Create a Schedule Apex

Schedule the Apex class:

1. From the Setup page search for "Apex Classes" in quick search.
2. Click on "Schedule Apex" as shown below.



3. Click on Schedule Apex and enter the Job name.
 - a. Job Name: DeleteAssignmentSchedule
 - b. Apex Class : ScheduleClass (from clicking on lookup icon)
 - c. Frequency: Monthly
 - d. Preferred Start Time: Select any time



4. Click Save.

Field service workorder optimization

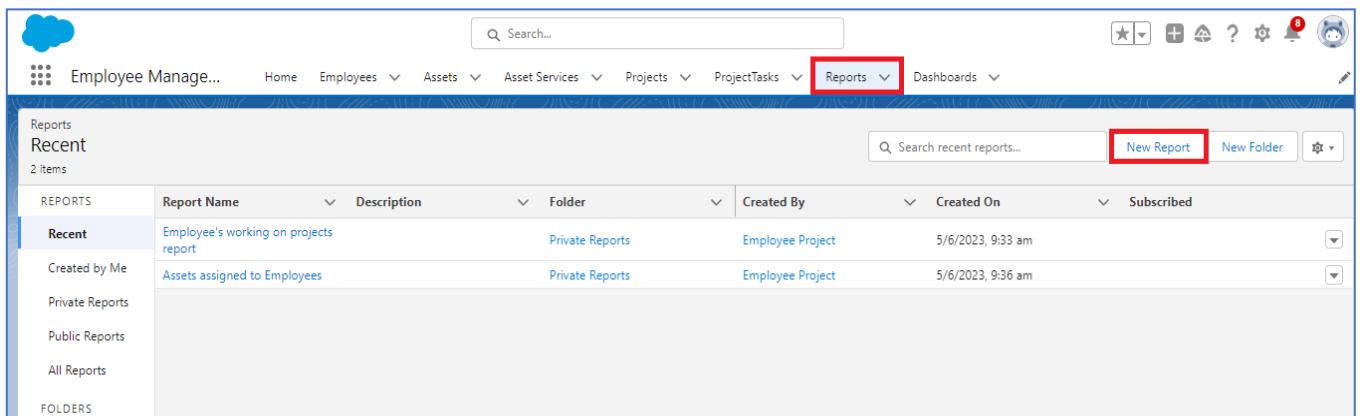
Milestone – 9

Reports & Dashboards

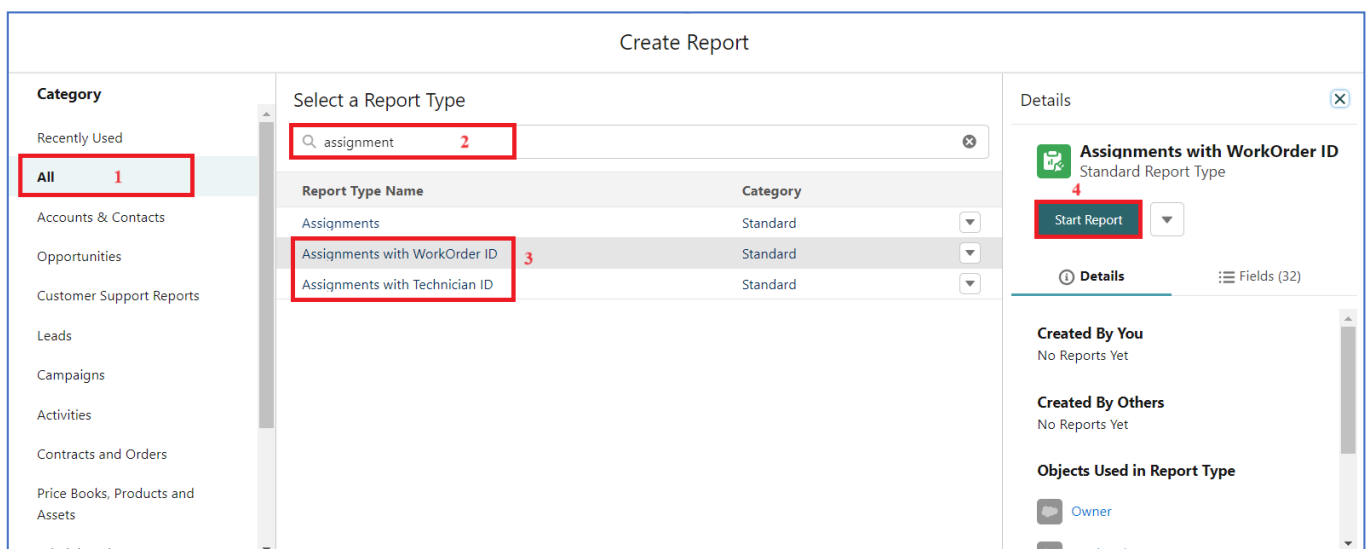
Salesforce Reports and Dashboards are powerful tools that empower users to visualize and analyze data within the Salesforce platform. They play a crucial role in providing insights, monitoring performance, and making informed business decisions.

Activity - 1: Report

1. Go to the app --> click on the reports tab
2. Click New Report.



3. Select report type from category or from report type panel or from search panel --> click on start report.



Field service workorder optimization

4. Customize your report

- Add fields from left pane as shown below
- Grouped by workorder ID

The screenshot shows the 'Employee Manage...' interface with the 'Reports' tab selected. A new report titled 'New Employees Report' is being configured. The left sidebar shows the 'Fields' pane with 'Groups' and 'Columns' sections. The 'Columns' section is expanded, showing a list of fields: 'Employee Employee Name', 'Employee ID', 'Reports to', 'Login Time', 'Logout Time', 'Mode of Work', and 'LinkedIn Profile'. The 'Add column...' button is highlighted with a red box. The main area shows a preview of the report with two columns: 'Employee Employee Name' and 'Employee ID'. The preview table has two rows: 'Employee' and 'Emp for Junction test'. The 'Save & Run' button is highlighted with a red box.

Employee Employee Name	Employee ID	Reports to	Login Time	Logout Time	Mode of Work	LinkedIn Profile
Employee	a05400000H1T0	-	-	-	-	https://www.linkedin.com
Emp for Junction test	a05400000H18VY	-	8:00 am	5:00 pm	-	https://www.linkedin.com

5. Save or run it.

Note: Reports may get varied from the above pictures as the data might be different.

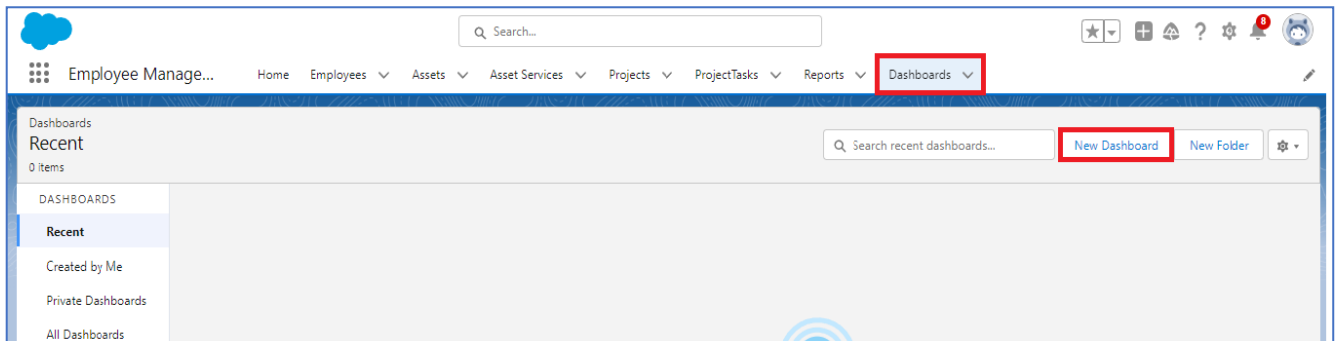
Field service workorder optimization

Activity - 2 : Create Reports

1. Create a report with report type: "WorkOrders Status Reports".
2. Create a report with report type: "Technician and Assignment Details Reports".

Activity - 3 : Dashboard

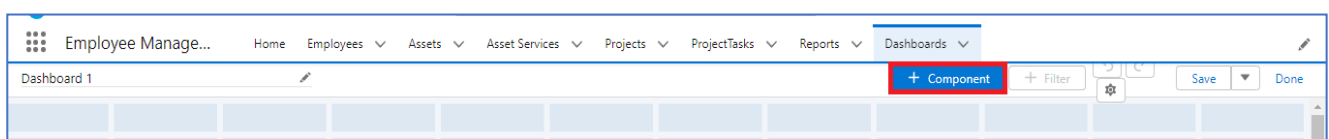
1. Go to the app --> click on the Dashboards tabs.



2. Give a Name and click on Create.

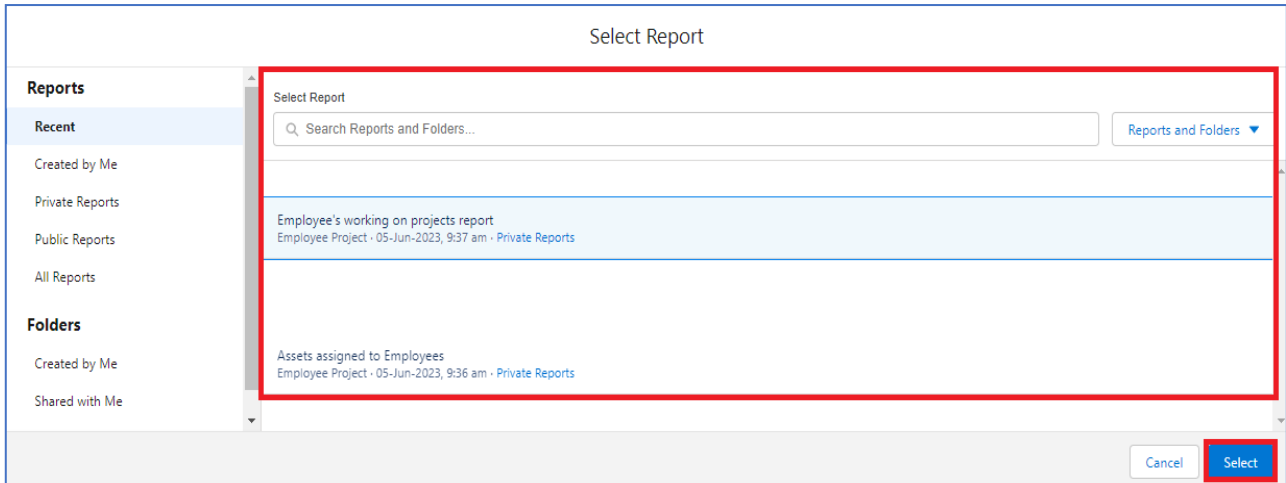
A screenshot of a 'New Dashboard' form. The form has three main sections: 'Name', 'Description', and 'Folder'. The 'Name' section has a text input field containing 'Dashboard 1'. The 'Description' section has an empty text input field. The 'Folder' section has a dropdown menu showing 'Private Dashboards' and a 'Select Folder' button. At the bottom right, there are two buttons: 'Cancel' and 'Create'. The 'Create' button is highlighted with a red box.

3. Select add component.



- 4 Select a Report which we have created in the previous activities and click on select.

Field service workorder optimization



5. Click Add then click on Save and then click on Done.

Activity - 4 : Create Dashboards

Create another Dashboard as we discussed in activity 3 which shows the details of completed workorder status in a vertical bar graph.

Field Service WorkOrder Optimization

GitHub Link:

<https://github.com/balub0115/field-service-workorder-optimization---LTVIP2025TMID29619>

Demo Video Link:

https://drive.google.com/file/d/13J3XoYS-szOgl7erpU8yfrdZxkFWApGf/view?usp=drive_link